

UNIT A

**The Nervous
and Endocrine
System**



Chapter 11

The Nervous System

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- Topic 1: Nerves & Action Potential
- Topic 2: Central & Peripheral Nervous System
- Topic 3: Sensory Reception

11.1 Nerves & Action Potential

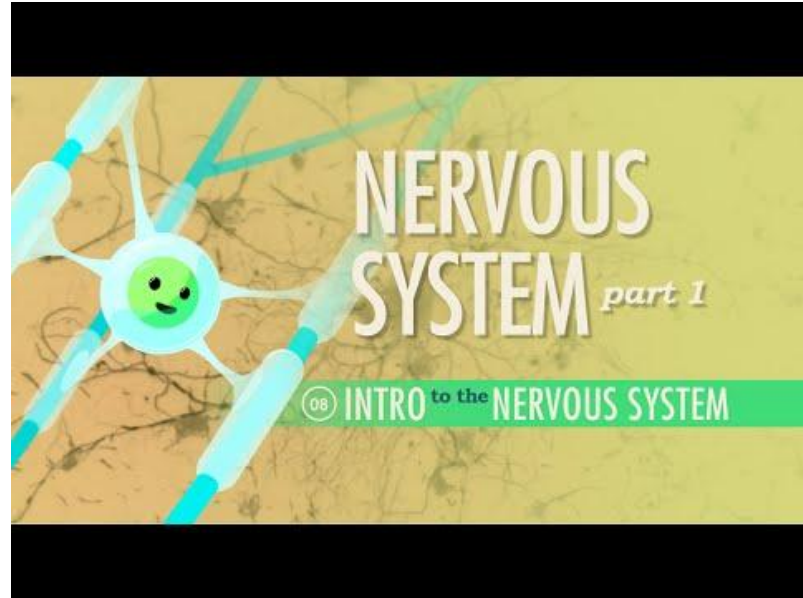
- Describe the general structure and function of a neuron and myelin sheath
- Explain the formation and transmission of an action potential
- Explain the intensity of a signal in terms of the all-or-none response
- Explain the transmission of a signal across a synapse, as well as the main neurotransmitters involved
- Describe the organization of neurons into nerves and the function of reflex arcs

*** Textbook pages 364-384**

Introduction

Watch the following video for a general overview of what you will be learning about.

Feel free to come back to this video later on to review as well.



Overview

The nervous system works in conjunction with the endocrine system to respond to both internal and external environmental change, thereby maintaining *homeostasis* within the body.

- The **nervous system** responds via **electrochemical messages** relayed from the brain.
- The **endocrine system** responds via chemical messengers relayed through the bloodstream (hormones).

Nerves

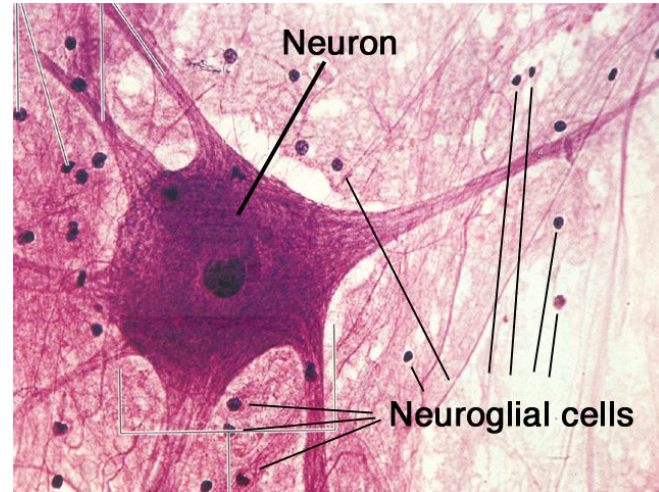
Two main types of nerve cells found in the nervous system:

1. Glial Cells

- do not conduct electrical signals
- used for structural and nutritional support

2. Neurons (nerve cells)

- conduct nerve impulses throughout the body
- supported by glial cells



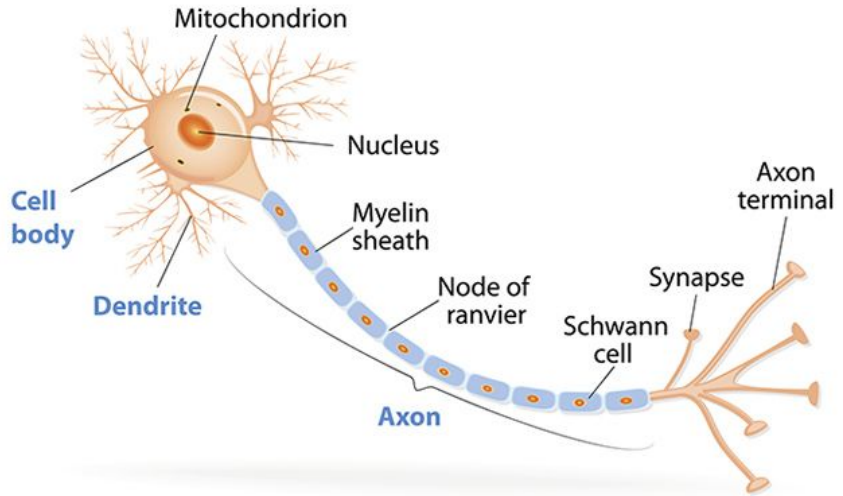
Neurons

Neurons are the basic unit of the nervous system...

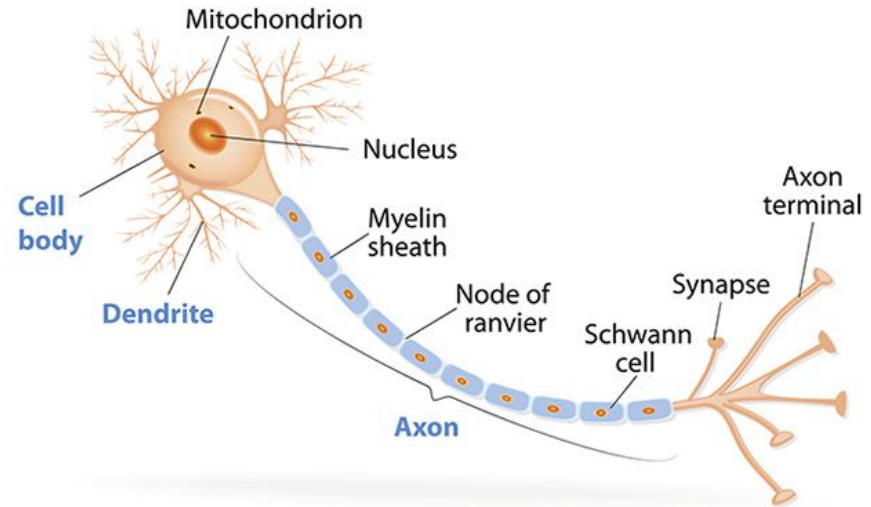


Neuron Structure

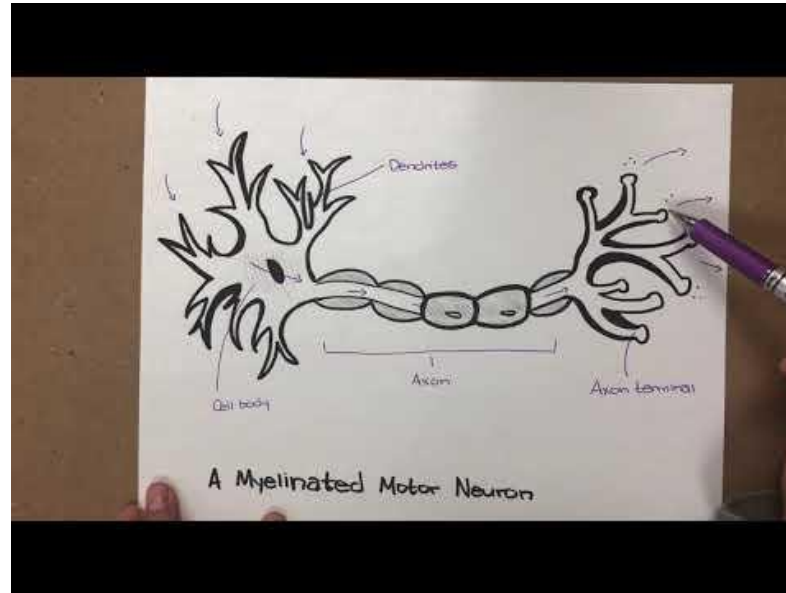
- **Dendrites** are short thread-like branches which act to accept nerve impulses from other neurons and carry them towards the **cell body**, which is the main part of the cell, containing the nucleus and other organelles
- **Axons** are longer branches which carry nerve impulses away from the cell body
- A **myelin sheath** surrounds many axons, insulating the neuron and speeding up the rate of impulse transmission.



- **Schwann Cells** , a type of glial cell, are responsible for producing the myelin that surrounds each axon.
- In myelinated axons, gaps between Schwann cells are referred to as the **nodes of Ranvier** . Electrical impulses “jump” from node to node, which is referred to as **saltatory conduction** . This speeds up transmission, avoiding having to travel down the entire length of the axon.
- Once the electrical signal reaches the **axon terminal** , it is then passed on to the dendrites of an adjoining neuron.
 - **Synapse**: the space/junction between two adjoining neurons



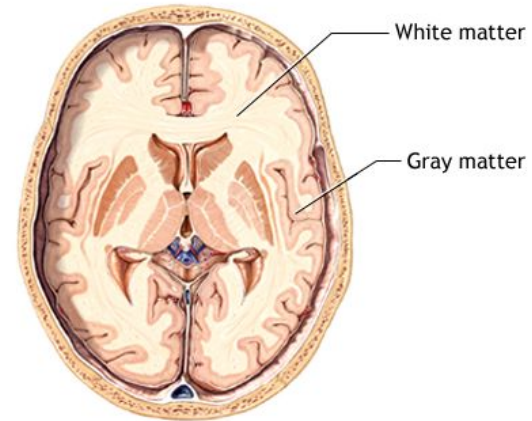
Re-cap: The Structure of a Neuron



Myelination

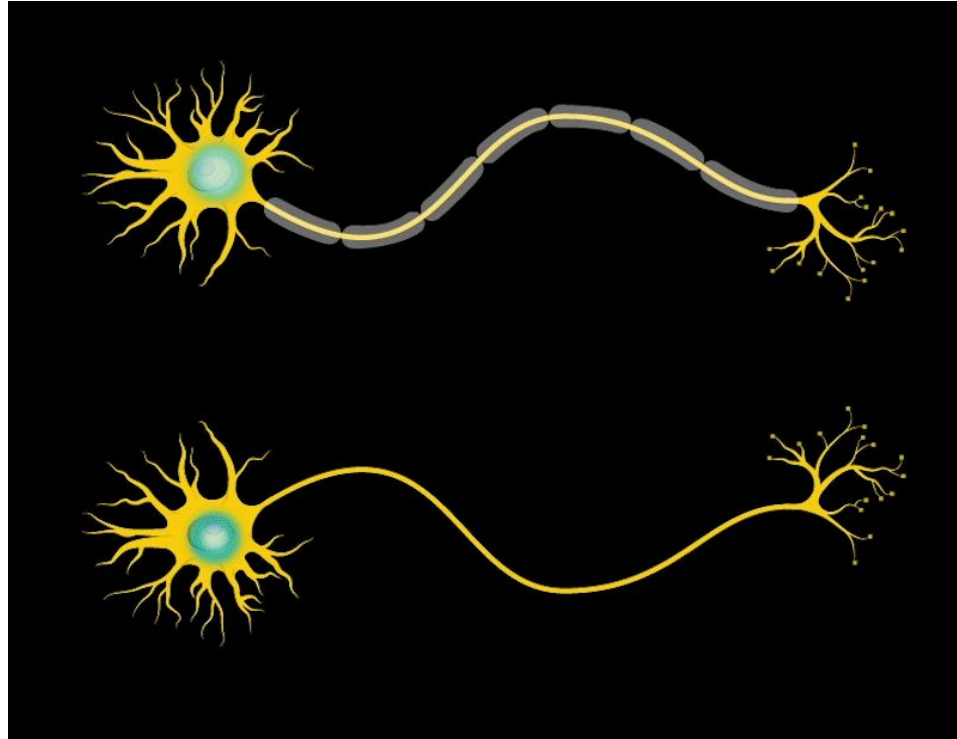
The myelination of neurons is vital for proper signal transduction within the nervous systems.

Myelinated Neurons	Unmyelinated Neurons
Increased rate of nerve impulse conduction (saltatory conduction)	Slower nerve impulse conduction (impulse has to travel the entire length of the axon)
make up the white matter of your brain, which is responsible for conducting nerve impulses.	make up the grey matter of your brain, which is responsible for processing information and generating nerve impulses.
capable of regenerating after injury	not capable of regenerating after injury



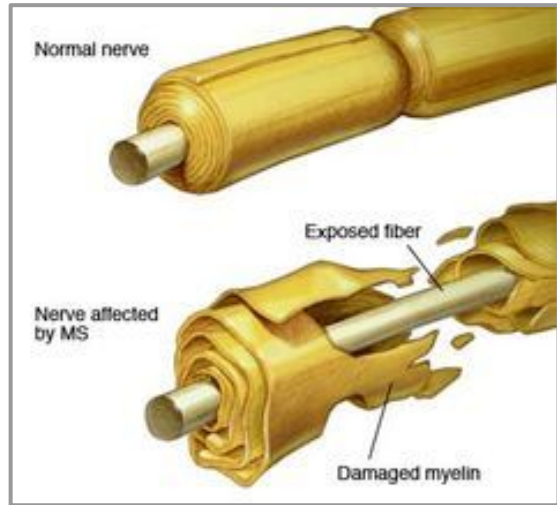
Myelination

Observe how the speed of the impulse is greater in the myelinated neuron.



Multiple Sclerosis

MS is a genetic disorder which involves damage to the myelin sheath



- Symptoms include numbness and tingling of the limbs, muscle spasms, and a loss of balance and coordination.
- The more scar tissue that builds up along the nerve, the worse the symptoms become. As a result, MS gradually worsens with age.

Types of Neurons

Three main types of neurons work together to signal *effectors* (e.g. muscle cells) to respond appropriately to different stimuli:

1. **Sensory Neurons** (*afferent*)

- *Gather information from sensory receptors* (i.e. touch/sight/sound/taste receptors) and transmit these impulses to the brain & spinal cord

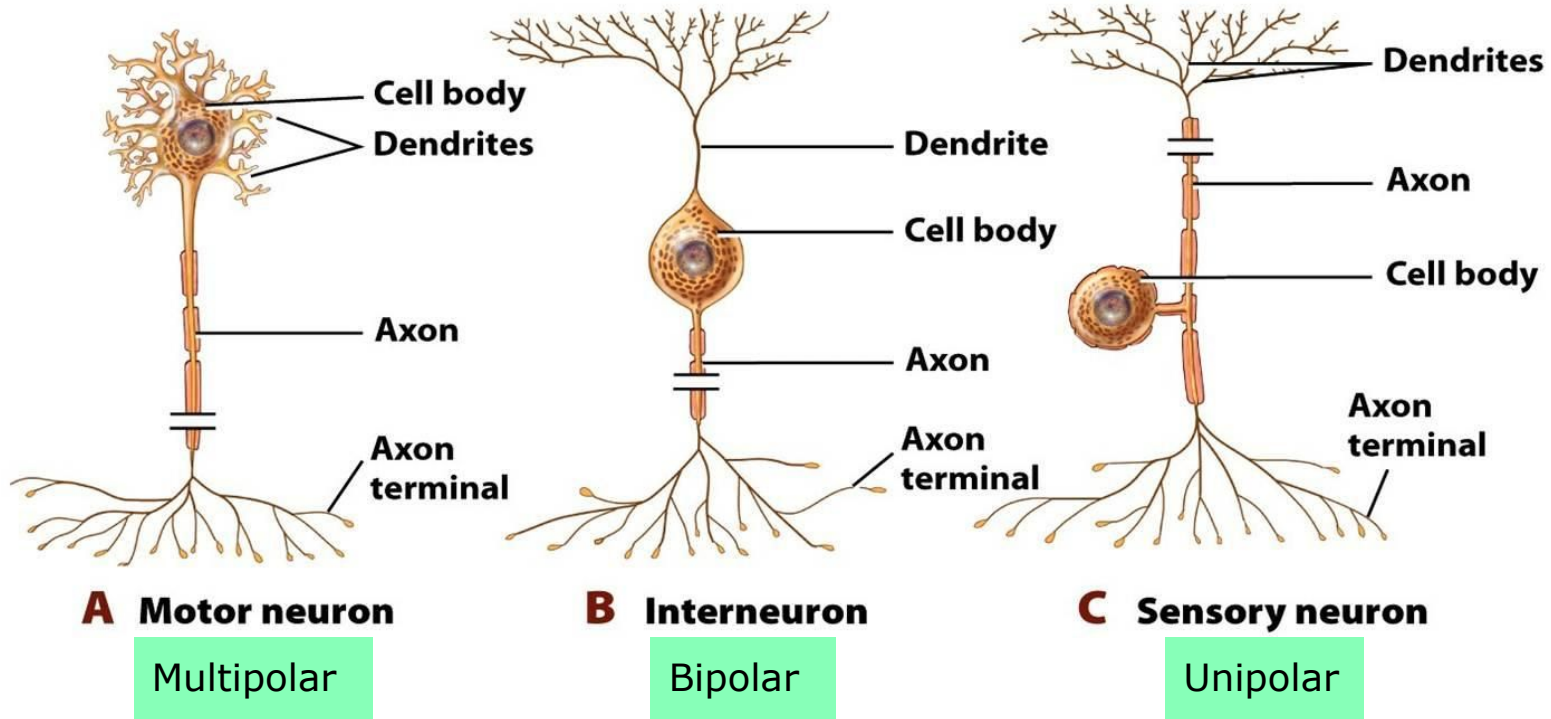
2. **Interneurons**

- Found in the Central Nervous System (brain and spinal cord)
- Process and integrate incoming sensory info from sensory neurons and relay outgoing information to motor neurons

3. **Motor Neurons** (*efferent*)

- *Transmit information to effectors* , such as muscles and glands

- Motor, inter, and sensory neurons can be further classified based on the number of branches formed from the cell body.



Typical Nerve Impulse Pathway

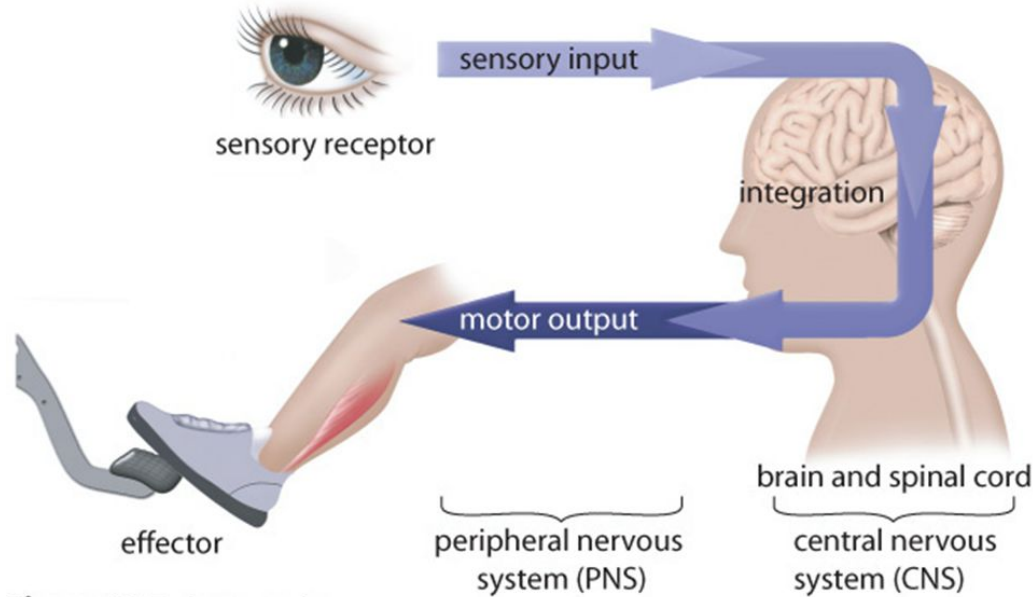
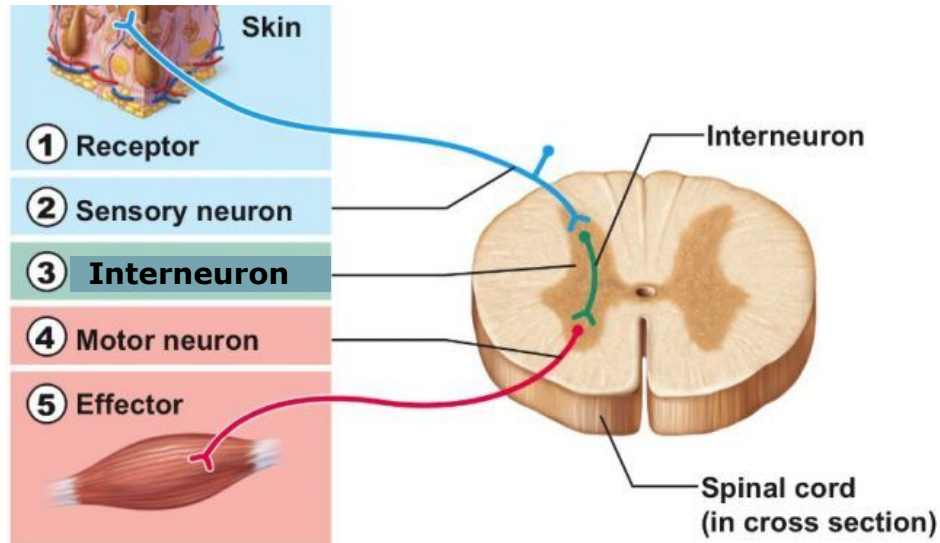


Figure 11.7 An overview of the neurons in a nerve impulse pathway

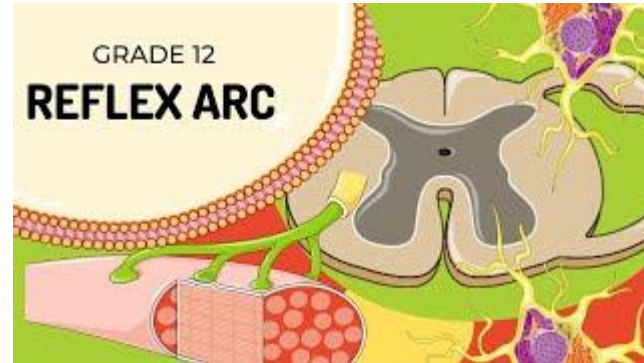
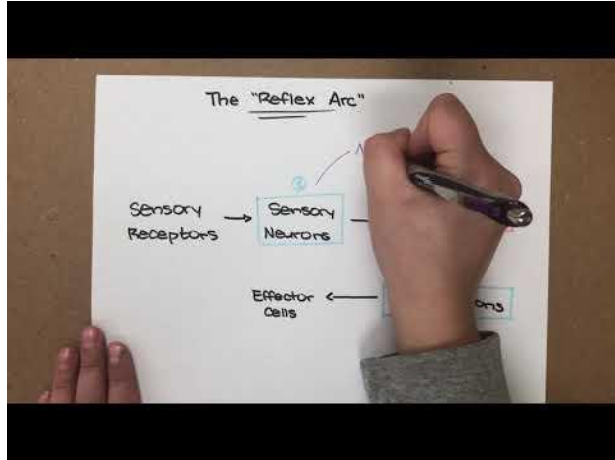
Neural Circuits and Reflex Arcs

Reflex arcs allow us to understand simple neural circuits typically composed of the five components below:



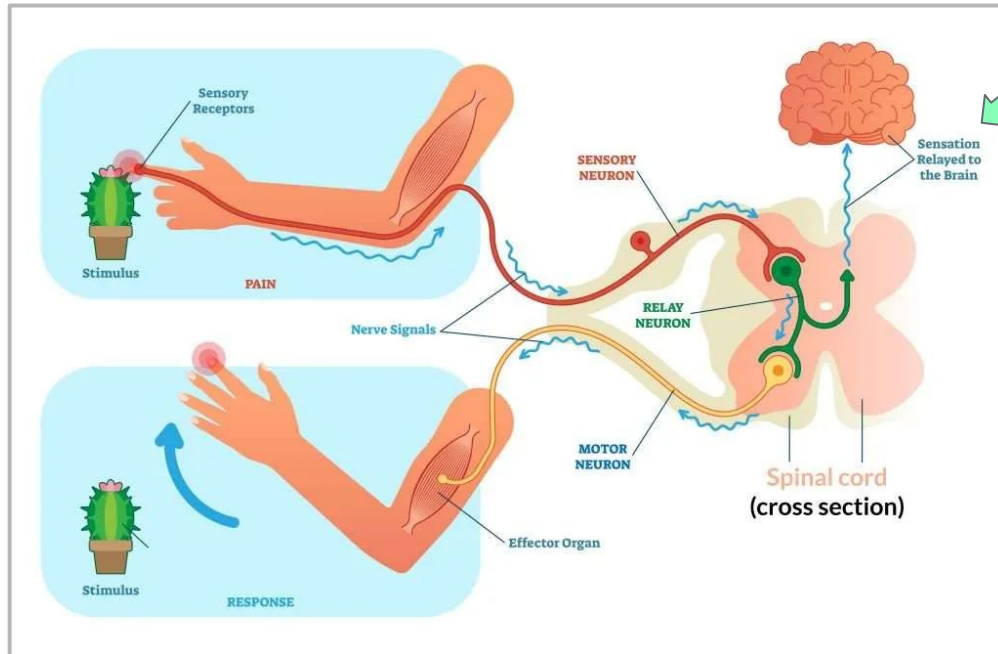
In some reflex arcs, the interneuron is not required, but there must be a minimum of a sensory and a motor neuron.

Reflex Arcs



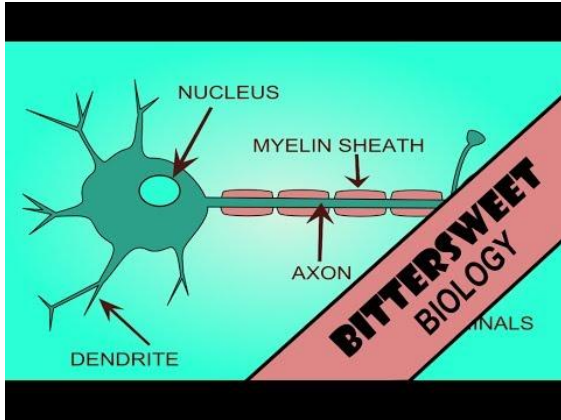
Reminder!

During a reflex arc, the information will still travel to the brain for further processing, however it first bypasses the brain and goes directly through the spinal cord to generate a quicker response.



Conduction of Nerve Impulses (Electrical Event)

A nerve impulse or ***action potential*** has both a chemical and electrical component (hence the term electrochemical impulse)



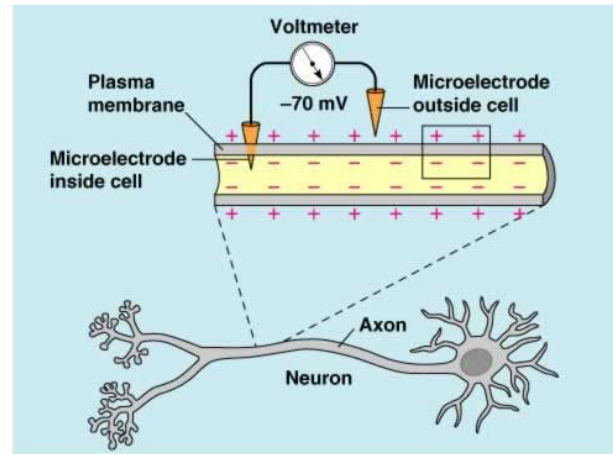
Four stages of a nerve impulse:

- 1) Polarized/resting state
- 2) Depolarization
- 3) Repolarization
- 4) Refractory Period

1) Resting State

The difference in charge across the membrane of a resting neuron is called *resting membrane potential*.

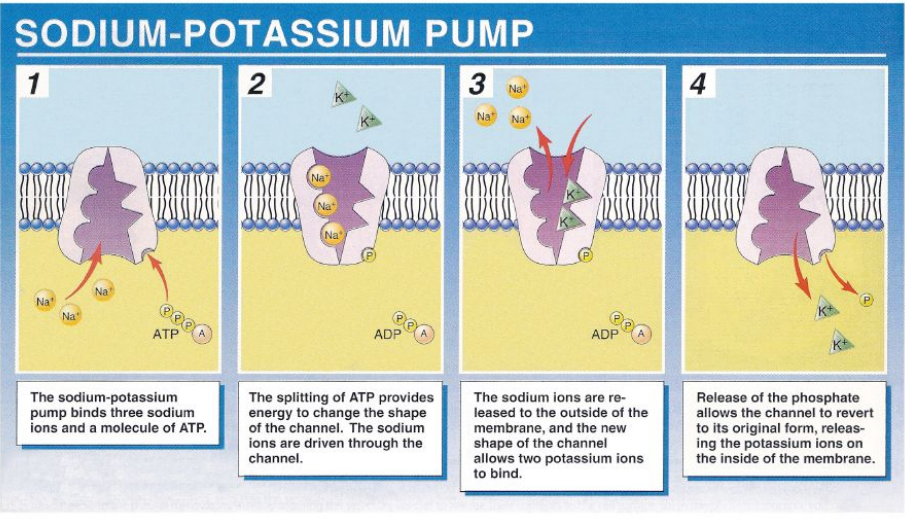
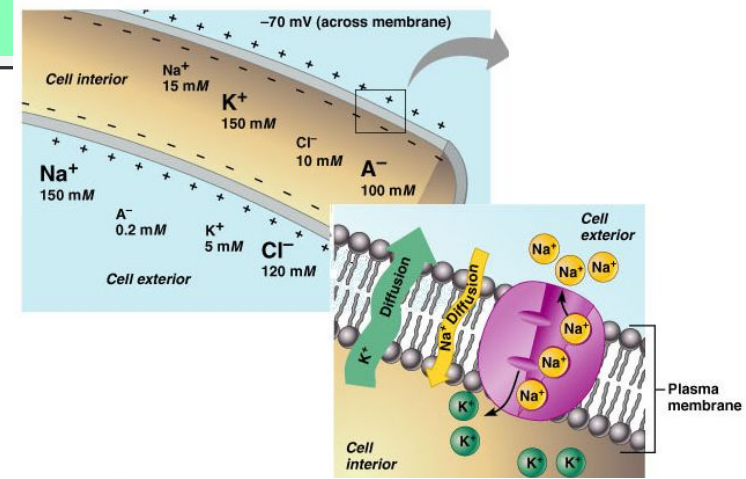
- The inside of a neuron has a slight negative charge at rest, whereas the outside has a slight positive charge.
- This results in a resting potential of -70 mV in most neurons
 - *Note: we often use specific numbers for voltages in our examples, but these can change from one neuron to another*



Resting State & the

Sodium-Potassium Pump

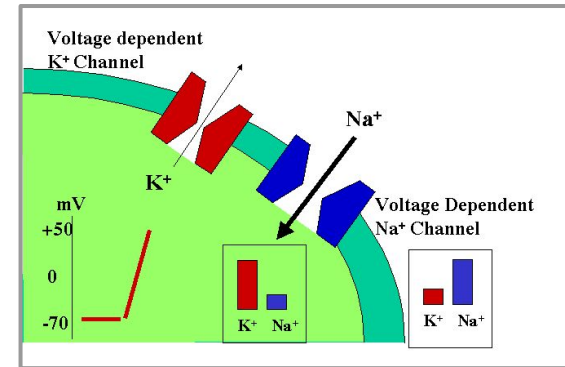
1. Inside the resting cell, there exists a higher concentration of potassium (K^+) than sodium (Na^+); outside the cell, the opposite is true.
2. Potassium ions are able to passively diffuse out of the cell more easily than sodium ions can diffuse into the (K^+ = "leaky" ion channels)
3. In an attempt to balance this difference, the sodium-potassium pump along the membrane surface will pump two K^+ back into the cell in exchange for three Na^+
4. As a result, as excess of positive charge accumulates outside of the cell membrane, causing the inside of the cell to become negatively charged relative to the outside. A constant membrane potential of -70 mV is maintained



2) Depolarization

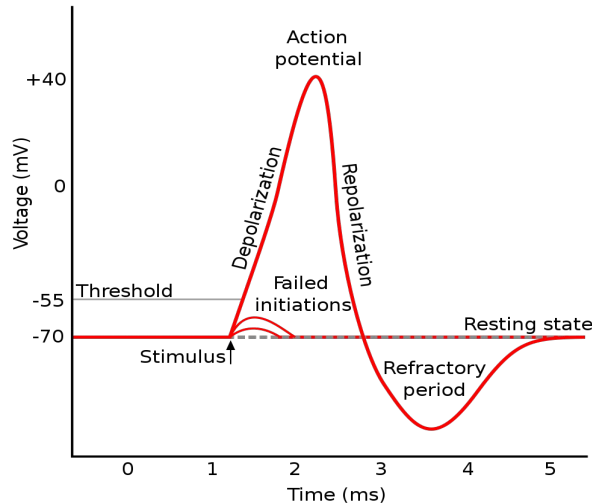
Action potential occurs when a neuron is stimulated by an electrical impulse.

1. If a strong enough impulse is received and the threshold potential is reached (approx. -55 mV), sodium gates will open, thus allowing Na^+ to diffuse freely across the membrane.
2. Na^+ rushes into the cell (down their concentration gradient $\rightarrow$ facilitated diffusion), leading to a slight positive charge on the inside relative to the outside.
3. This reverses the membrane potential from -70 mV to $+40$ mV; the membrane is now said to be *depolarized*.
4. Once Na^+ reaches equilibrium, the membrane once again becomes impermeable to them, and the gates close.



Depolarization & Threshold Potential

An action potential is considered to be an “**all-or-none**” event because any stimulus that fails to achieve the threshold membrane potential of at least -55 mV will have no effect

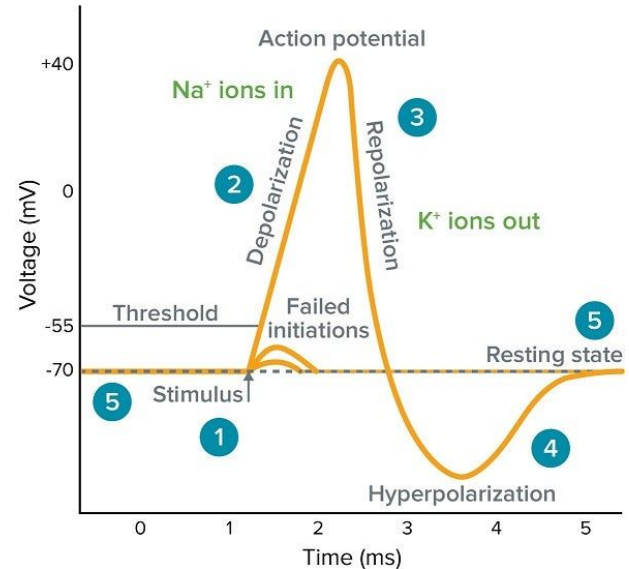


- Note that increasing the stimulus strength does not increase the impulse strength; a neuron will either fire or not fire
- The **intensity of a stimulus is instead experienced as an increased frequency of nerve impulses** (i.e. a more intense stimulus will simply cause more neurons to depolarize)

3) Repolarization

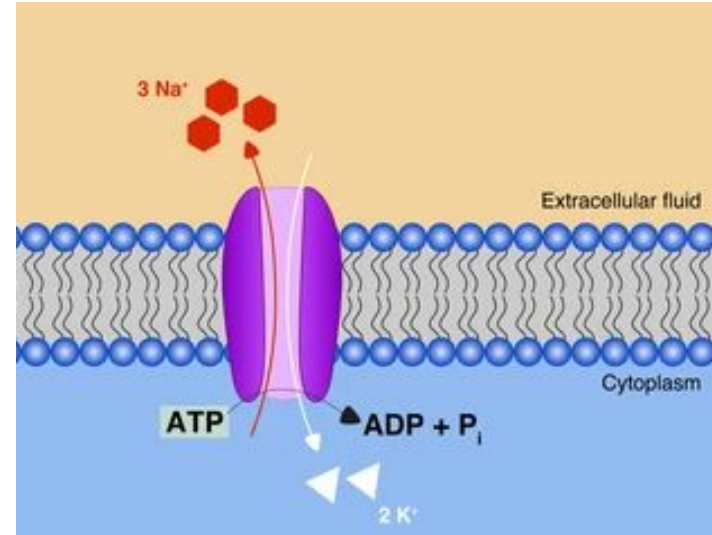
Once the action potential has peaked, Na^+ gates close, and K^+ gates open.

- This restores the positive charge on the outside of the membrane; however, Na^+ and K^+ concentrations are briefly reversed compared to at the resting state (Na^+ is higher inside the membrane)
- The K^+ channels are slow to close, which results in the **hyperpolarization** of the neuron. This means that the inside becomes slightly more negative than the resting potential.

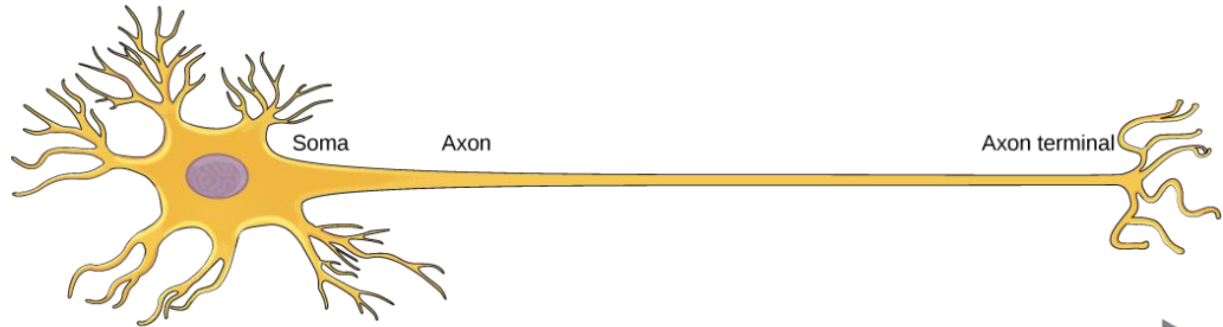


Na⁺/K⁺ exchange pump helps reset the correct distribution of ions

- Even though during repolarization the inside becomes more negative again compared to the outside, we do not have the correct distribution of Na⁺ and K⁺ across the membrane. The sodium-potassium exchange pump helps reset the original distribution by pumping 3 Na⁺ out for every 2 K⁺ it pumps in.



Nerve transmission occurs in a wave-like pattern down the length of the axon.

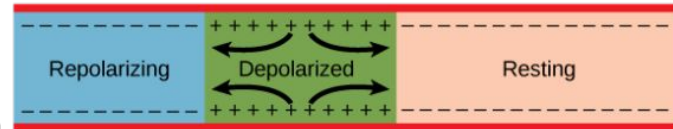


Direction of travel of action potential

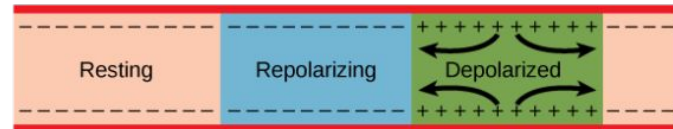
- a. In response to a signal, the soma end of the axon becomes depolarized.



- b. The depolarization spreads down the axon. Meanwhile, the first part of the membrane repolarizes. Because Na^+ channels are inactivated and additional K^+ channels have opened, the membrane cannot depolarize again.



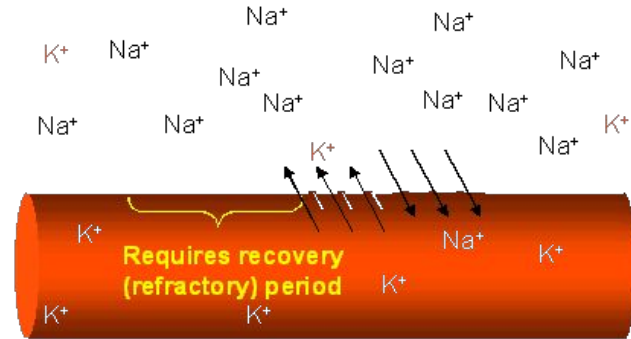
- c. The action potential continues to travel down the axon.



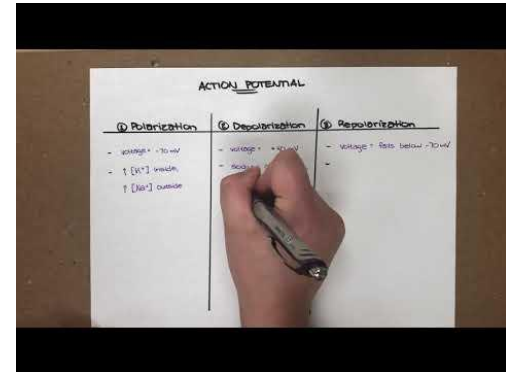
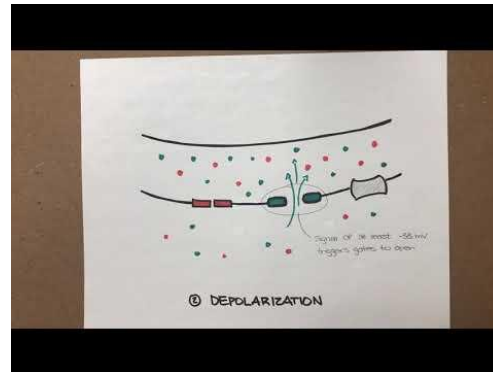
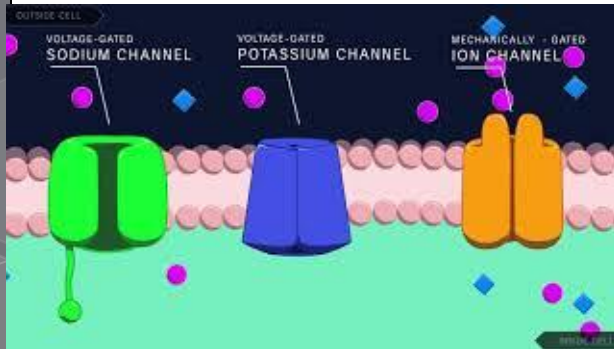
4) Refractory Period

Until the resting potential of the neuron has been properly restored, a second action potential cannot be conducted along the axon; this is referred to as the *refractory period*.

- During this time, the membrane cannot be made permeable to Na^+ , so a second wave of depolarization cannot occur.



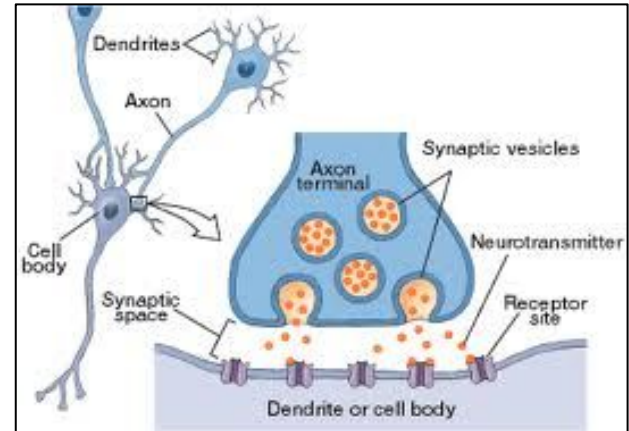
Re-cap: Action Potential



This is a tough topic... I would watch all three videos to make sure you really understand each step.

Signal Transduction across a Synapse (Chemical Event)

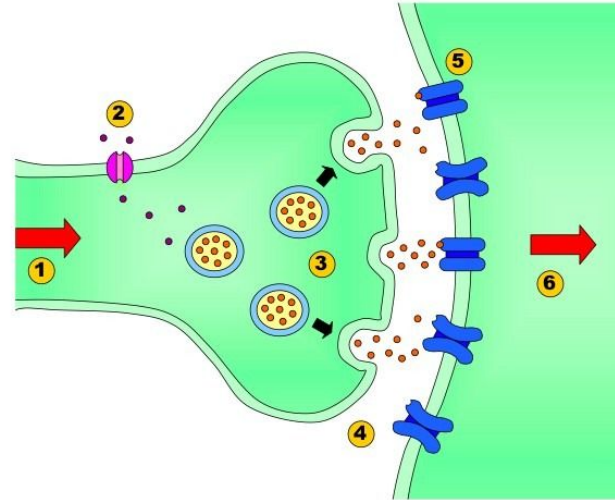
- Once the impulse has traveled down the length of the axon, it reaches the *axon terminal*.
- The axon terminal is in close contact with the *dendrites* of another neuron.
- For the signal to move to the next neuron, it must cross the space between the axon terminal and the dendrites of the subsequent cell; this space is referred to as the *synapse*.



Neurotransmitter Release

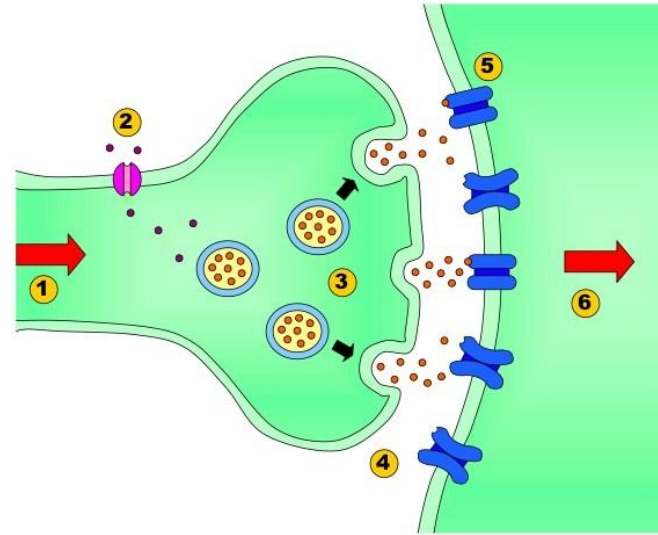
Neurotransmitters are required to help transmit the electrical impulse from one neuron to another. These are stored in vesicles in the axon bulb.

1. An action potential reaches the axon terminal
2. Calcium channel “gates” within the axon terminal open, causing calcium ions to flow into the cell
3. Calcium ions trigger the movement and fusing of the vesicles with the **presynaptic membrane** (exocytosis). In turn, neurotransmitters are released into the synapse.
4. Neurotransmitters **diffuse** across the synaptic cleft to the **post-synaptic membrane**



Neurotransmitter Binding

5. Once neurotransmitters have reached the post-synaptic membrane, they bind with receptors embedded in the membrane
6. If the neurotransmitter is excitatory, binding induces an action potential in the postsynaptic neuron
7. Neurotransmitters are then released by the receptors and either return to the presynaptic neuron or are broken down by enzymes such as cholinesterase on the post-synaptic membrane



Neurotransmitter Types

There are two main types of neurotransmitters:

1. **Excitatory** (e.g. glutamate, acetylcholine, epinephrine)
 - cause the sodium channels of the post-synaptic membrane to open and sodium to flow in, resulting in slight depolarization of the membrane and **lowering the threshold potential** of the post-synaptic neuron
 - e.g. acetylcholine is a neurotransmitter found in muscle cells; it excites the muscle membrane, causing depolarization and contraction of the muscle fibre
 - **Cholinesterase** (an enzyme) is required to break down acetylcholine after the action potential has occurred
 - Often when more than one neuron fires at the same time, a sufficient amount of neurotransmitter is released to cause depolarization of the postsynaptic membrane. The production of an action potential as a result is referred to as **summation**

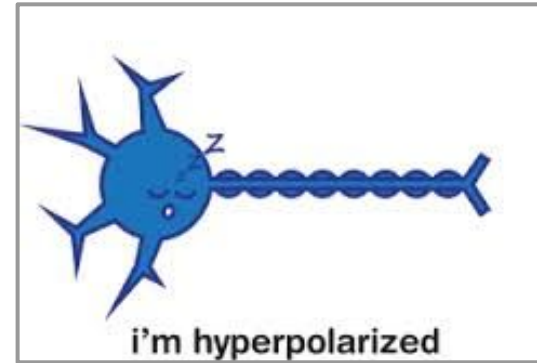


...insecticides and sarin nerve gas work by blocking the release of cholinesterase, causing muscles to remain in a state of constant contraction

Neurotransmitter Types Cont'd

2. Inhibitory (e.g. GABA)

- Trigger potassium channels to open, causing potassium ions to flow out and thus lowering the membrane potential.
- This leads to **hyperpolarization**, making it more difficult to generate an action potential (because the threshold level has been increased)
- Problems with GABA are often associated with epilepsy and Huntington's disease

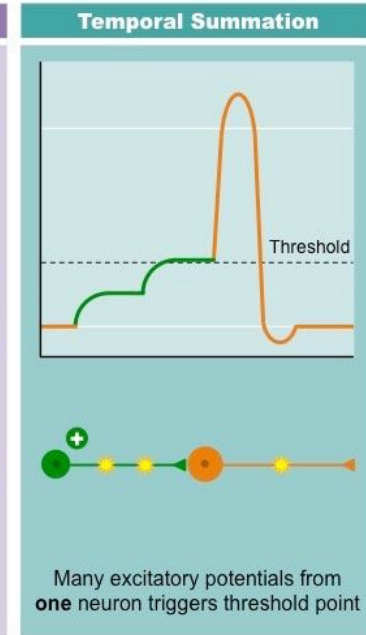
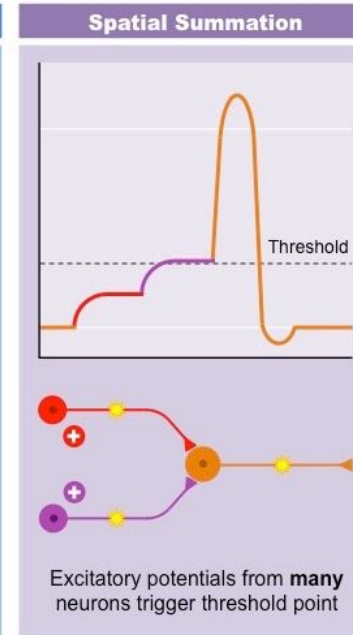
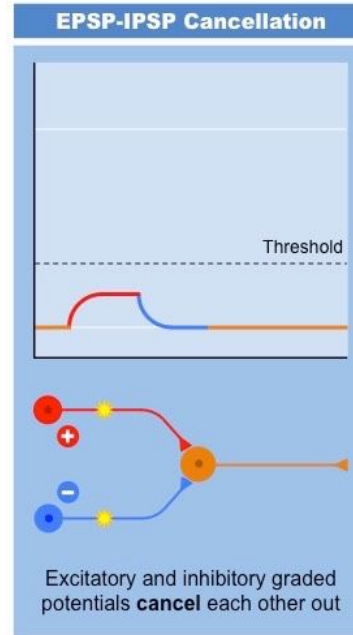


Excitatory vs. Inhibitory Neurotransmitters

Excitatory	Inhibitory
depolarize membranes by increasing permeability to Na^+	hyperpolarize membranes by increasing permeability to K^+ or Cl^-
move membrane potential towards threshold (causes it to become more positive)	move membrane potential away from threshold (causes it to become more negative)

Summation

- Summation is the process that determines whether or not an action potential will be generated by the combined effects of excitatory and inhibitory signals (threshold potential may or may not be reached).
 - Spatial summation:** from multiple simultaneous inputs
 - Temporal summation:** repeated inputs

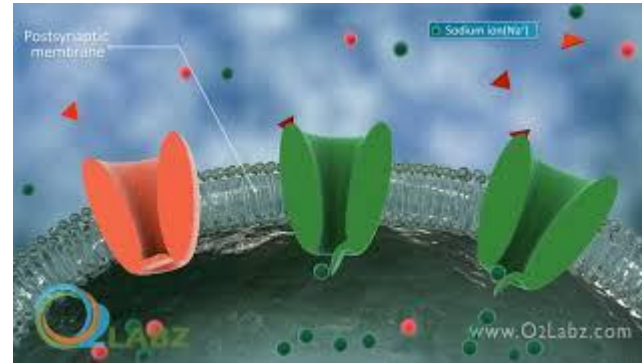
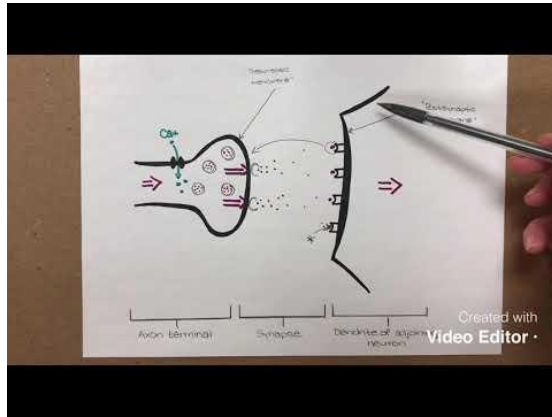


Some Common Neurotransmitters:

*Memorize

Neurotransmitter	Action	Secretion Sites	Major Effects
acetylcholine*	Excitatory	Neuromuscular regions, CNS, PNS	Skeletal muscle contraction/parasympathetic
norepinephrine*	Excitatory	CNS, PNS	Wakefulness/sympathetic
dopamine	Generally excitatory	CNS, PNS	Voluntary movement and emotion
serotonin	Generally inhibitory	CNS	Sleep
GABA*	Inhibitory	CNS	Motor behaviour

Re-cap: Synaptic Transmission



Drugs and the Nervous System

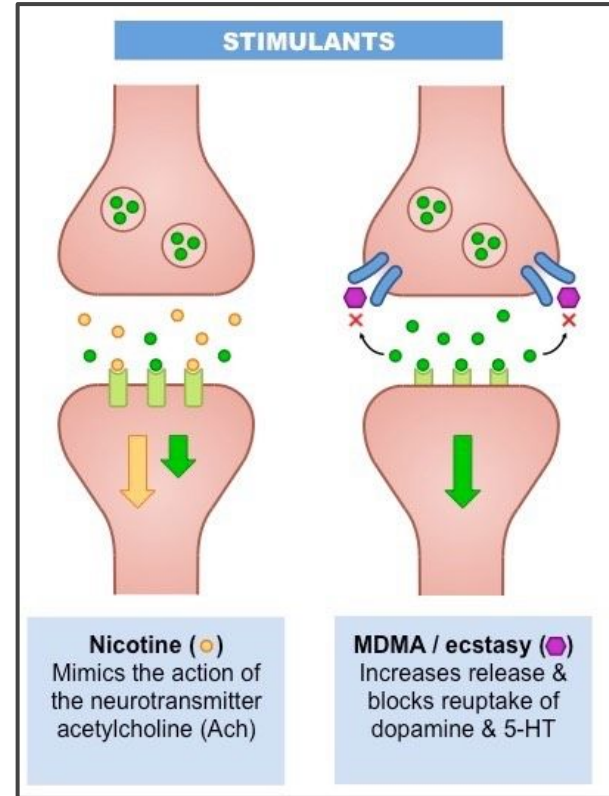
- You do not need to memorize information in these next two slides. However, make sure you understand how drugs are able to alter synaptic transmission using these mechanisms.
- DO, however, play around with the following simulation. The best way to understand many biological systems is to observe what happens when they are not functioning normally.
<https://learn.genetics.utah.edu/content/addiction/mouse/>



Action of Drugs at the Synapse

Mechanisms of stimulants

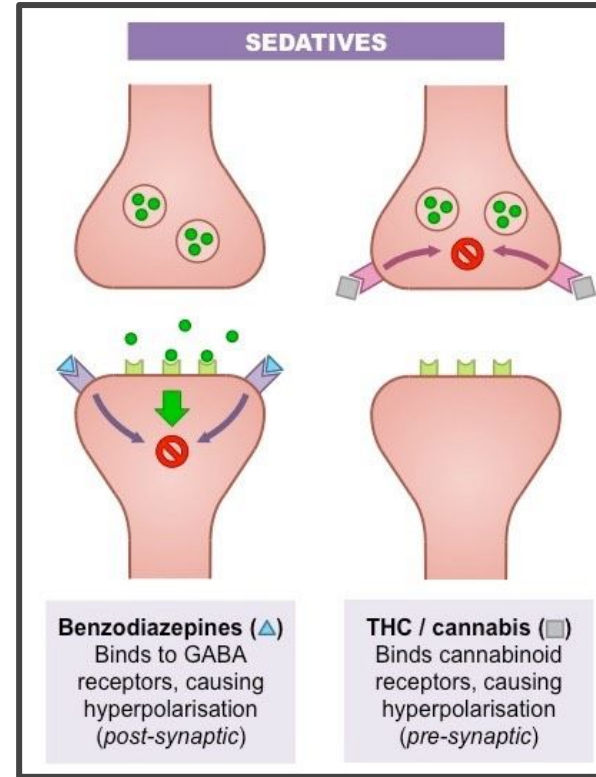
- 1) Drug mimics the neurotransmitter & stimulates the receptor.
- 2) Drug decreases the rate of neurotransmitter breakdown or re-uptake, leading to continued stimulation of the receptor.
- 3) Drug increases the rate of neurotransmitter release from the presynaptic neuron.



Action of Drugs at the Synapse

Mechanisms of depressants:

- 1) Drug blocks the receptor site so that the neurotransmitter cannot interact with the receptor.
- 2) Drug decreases synthesis & storage of a neurotransmitter at the presynaptic neuron.
- 3) Drug increases rate of breakdown of a neurotransmitter in the synapse.

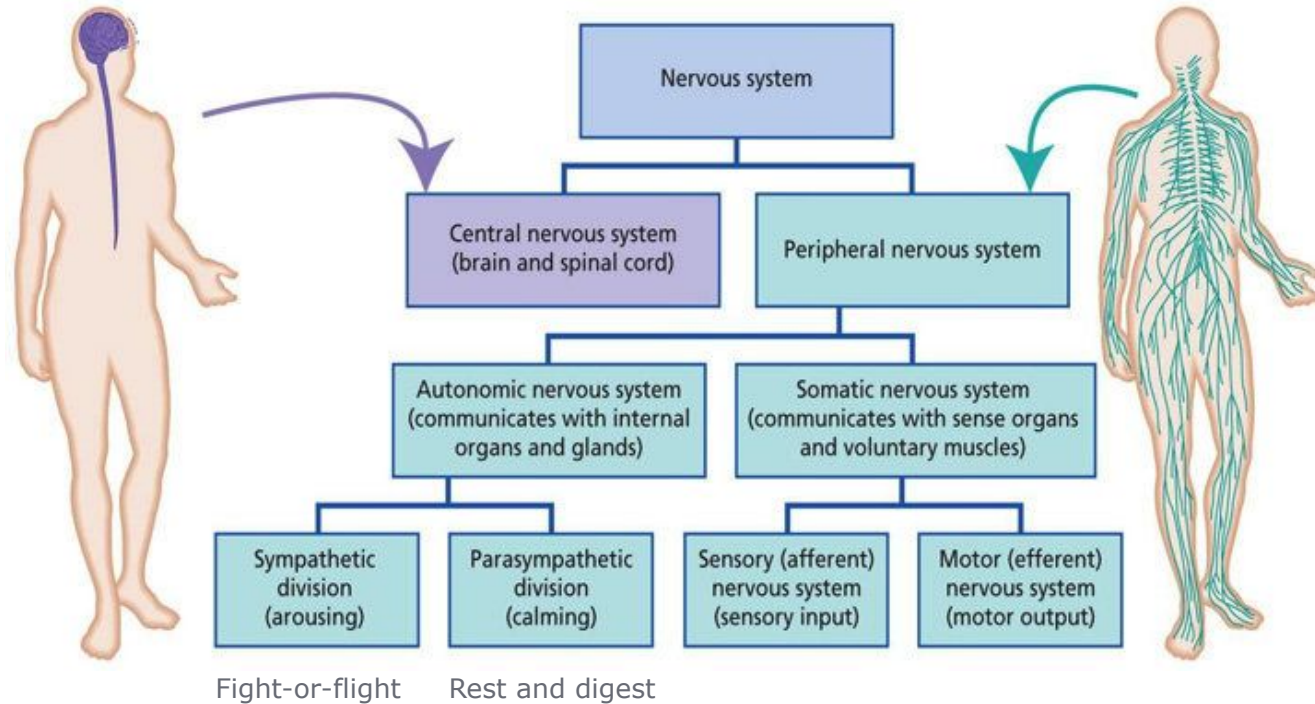


11.2 & 11.3 Central & Peripheral Nervous System

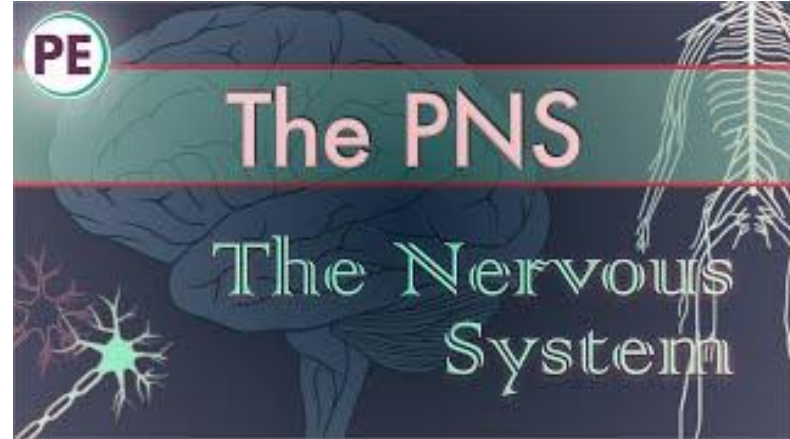
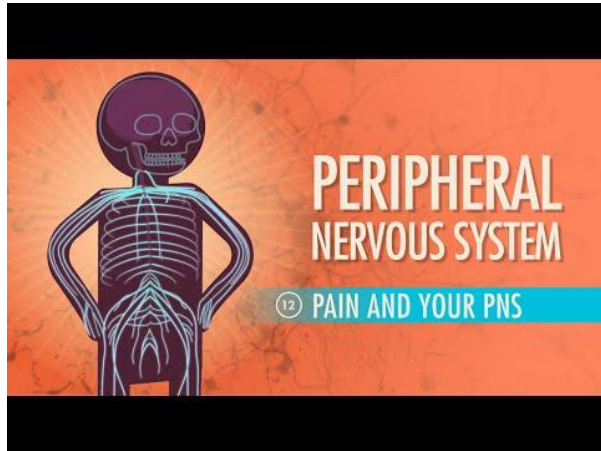
- Identify the principal structures & functions of the central and peripheral nervous systems (e.g. cerebral hemispheres and lobes, cerebellum, pons, medulla oblongata, hypothalamus, spinal cord, sympathetic and parasympathetic systems, sensory-somatic system)
- Explain the functions of the central and peripheral nervous systems in regulating the voluntary (somatic) and involuntary (autonomic) systems

*** Textbook pages 385-401**

Nervous System Organization



Peripheral Nervous System

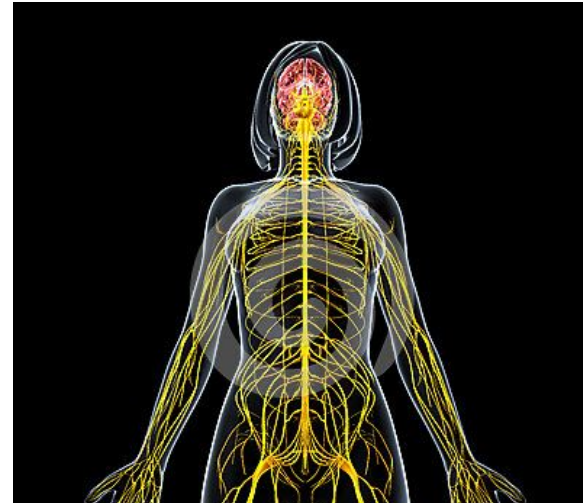


Peripheral Nervous System

The PNS consists of nerves which link the brain and spinal cord to the rest of the body, including sensory organs, muscles, glands, and internal organs.

Two main divisions:

- The somatic system
 - voluntary control
 - carries info from sensory receptors to skeletal muscles using sensory and motor neurons
- The autonomic system
 - involuntary control
 - maintains homeostasis (breathing, heart rate)



Subdivisions of the autonomic system:

1) Sympathetic response (stress response)

- “fight or flight” response
- activated during **stressful situations**
- the neurotransmitters acetylcholine & norepinephrine are released to excite muscles
- Pupils dilate, blood pressure increases & heart beats faster to prepare your body
- digestion slows and the bladder is constricted

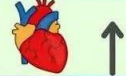
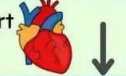
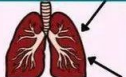
2) Parasympathetic response

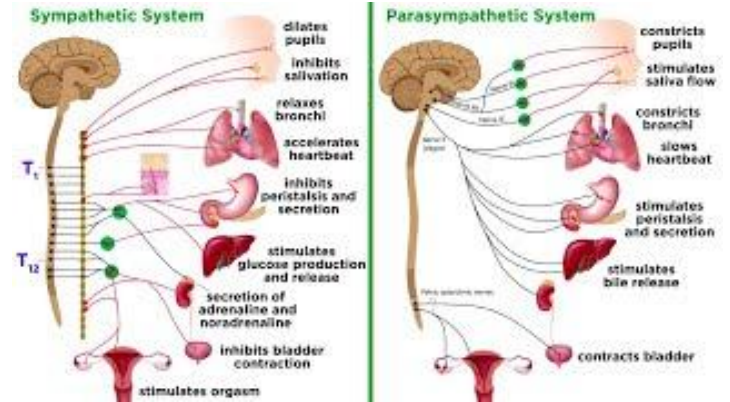
- “rest and digest” response
- activated when the body is **calm and at rest**
- associated with the neurotransmitter acetylcholine
- Pupils constrict, slows heart rate, reduces blood pressure, promotes digestion

many of the physiological changes in response to stress are detectable by polygraphs; heart rate, blood pressure, and breathing rate all tend to increase when someone is telling a lie



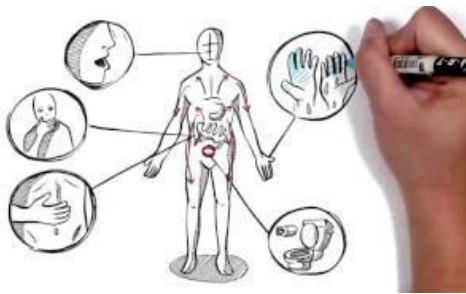
Sympathetic vs Parasympathetic

Body Organ /System	Sympathetic	Parasympathetic
Pupil	Pupil Dilation 	Pupil Constriction 
Salivary Gland	Decreased Saliva Production 	Increased Saliva Production 
Nasal Mucosa	Decreased Mucus Production 	Increased Mucus Production 
Heart Rate	Increased Heart Rate 	Decreased Heart Rate 
Arteries	Constriction 	Dilation 
Lungs	Bronchial Relaxation 	Bronchial Contraction 
GI System	Decreased Motility 	Increased Motility 
Kidney	Decreased Urine 	Increased Urine 
Sweat	Increased Sweating 	Normal <i>Cool!</i> 
Liver	Glycogen to glucose conversion 	Glycogen synthesis 



Extra: The Stress Response and Anxiety

The stress response, which triggers when our bodies perceive a threat, served an important evolutionary purpose in helping our ancestors survive dangerous situations. However, in modern times, stressors often aren't physical threats but rather psychological ones like exams, social pressures, or work deadlines. While the stress response can still help us cope with these challenges, if it's triggered too frequently or intensely, it can lead to chronic stress and anxiety. This is because our bodies aren't designed to handle prolonged periods of stress, and the constant activation of the stress response can have negative effects on both our physical and mental health, contributing to conditions like anxiety disorders.



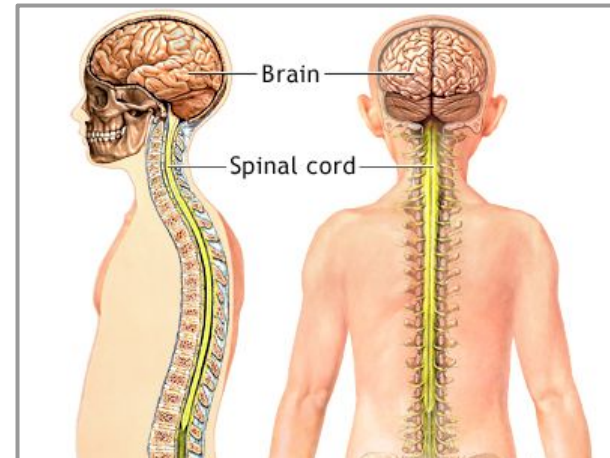
Some strategies to help reduce anxiety based on our understanding of the autonomic nervous system.



Central Nervous System

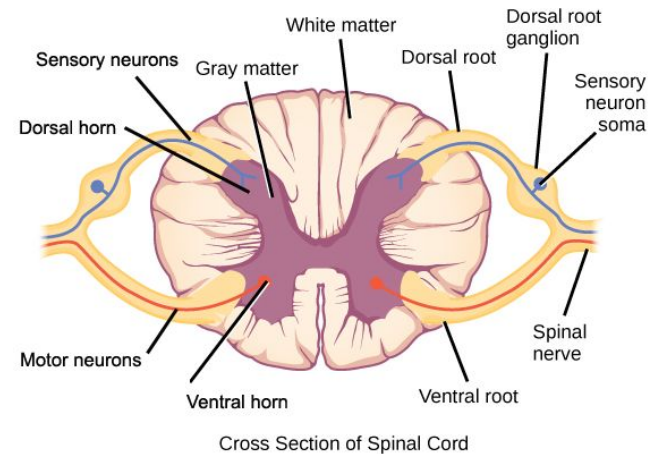
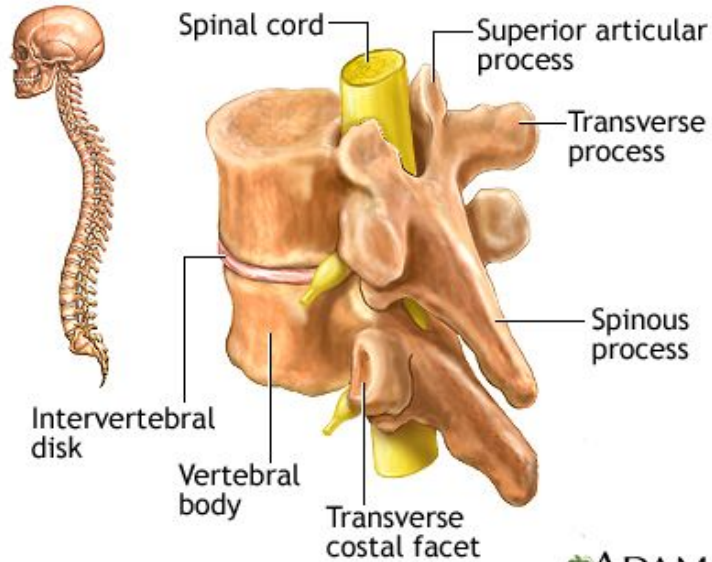
Composed of the **brain and spinal cord** , the central nervous system is the site of neural integration and processing.

- Damage to the CNS can affect everything from temperament and motor control to the maintenance of homeostasis depending on which region is harmed.
- The CNS consists of two types of nervous tissue: **grey matter (unmyelinated neurons)** & **white matter (myelinated neurons)**

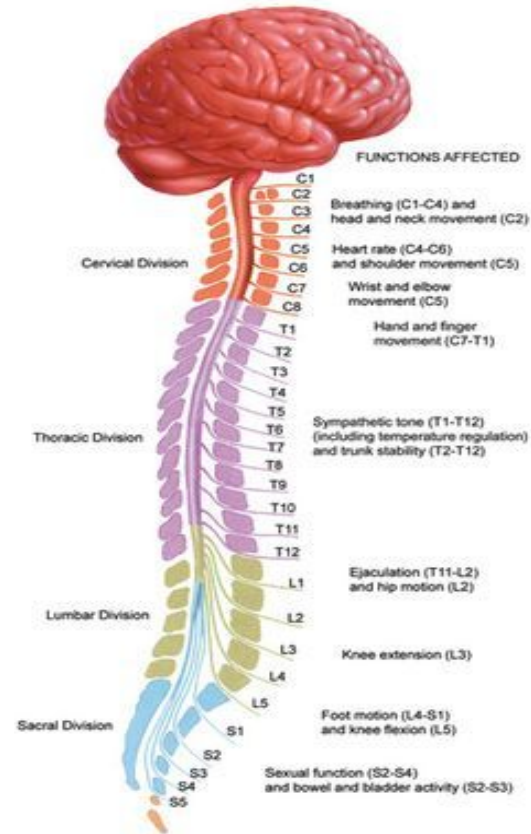


The Spinal Cord

The spinal cord provides a vital link for communication between the brain and peripheral nervous system...



- Each segment of the spinal cord is responsible for specific functions.
- The delicate tissues of the spinal cord are **protected by cerebrospinal fluid** and a series of vertebrae.
- Injury to areas of the spinal column closer to the brain results in the loss of more functions compared to injury of areas closer to the abdomen.



Extra: Levels of Paralysis

A spinal cord injury usually results in the area of the body below the injury site being partially or completely unable to communicate with the brain.

- **Paraplegia:** If the injury occurs below the first thoracic vertebrae in the upper back region, the paralysis is likely to be confined to the legs. A paraplegic patient will retain full use of their arms and hands.
- **Quadriplegia:** If the injury occurs above the first thoracic vertebrae in the cervical vertebrae or neck region of the body, the paralysis generally results in loss of movement in all limbs. Abdominal and chest muscles may also be affected, resulting in weakened breathing and the inability to properly cough to clear the chest.

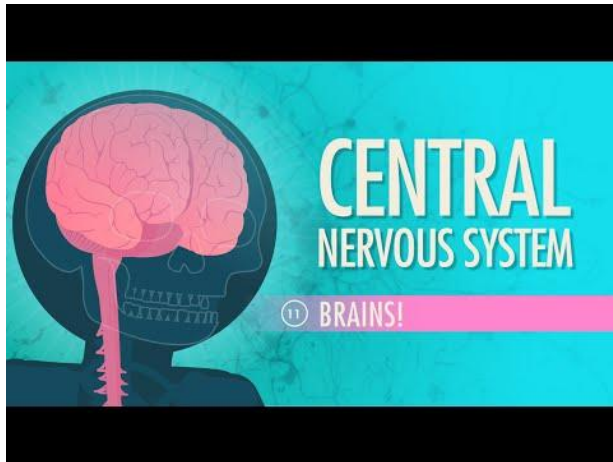


Paraplegic patients are not necessarily confined to a wheelchair; modern prosthetics and stem cell treatment can provide an increased level of mobility

The Brain

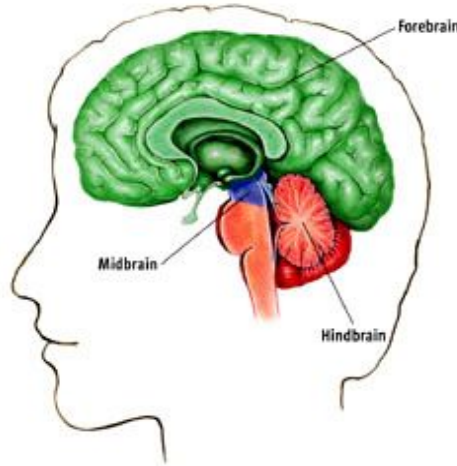
The brain is the center for intelligence, consciousness, and emotion. It also regulates bodily functions without conscious thought (e.g. breathing).

Brain intro:



- Enclosed within the skull, the brain is surrounded by a tough three-layer membrane known as the *meninges*.
- **Cerebrospinal fluid** circulates between the meninges and the spinal cord, acting as a shock absorber and a transport medium, carrying nutrients & waste to and from brain cells.

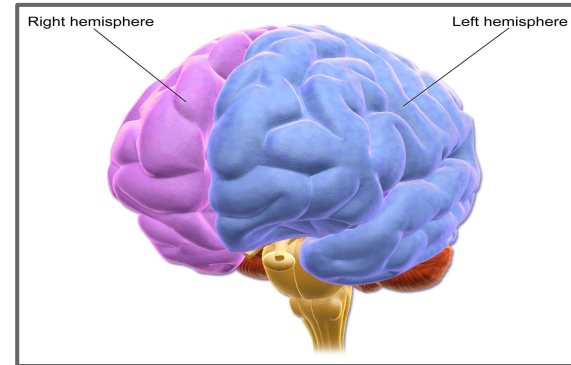
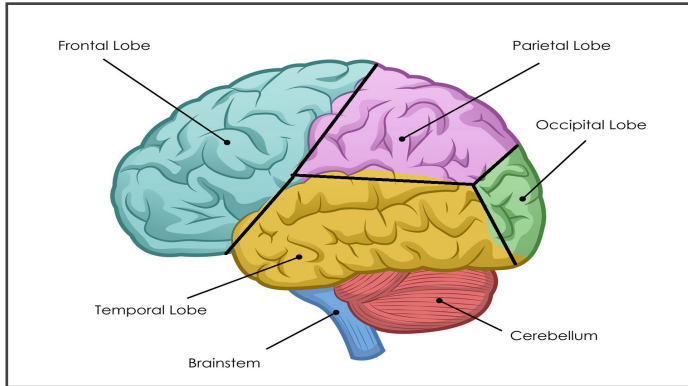
General Organization of the Brain



	FOREBRAIN	MIDBRAIN	HINDBRAIN
Main Structures	cerebrum basal ganglia hippocampus amygdala thalamus hypothalamus	tectum tegmentum	pons cerebellum medulla
General Roles	Learning & memory, conscious thought, problem solving, coordination & voluntary actions	Transmits signals between the forebrain & hindbrain; associated with vision, hearing, temp. Regulation & alertness	The “primitive” brain; controls breathing, heart rate, blood pressure, sleep/wake cycles, and balance.

The Forebrain: Cerebrum

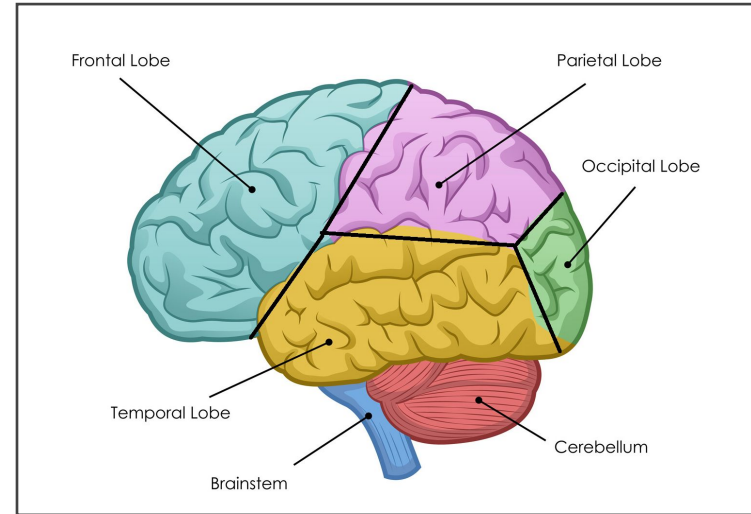
- The forebrain or *cerebrum* is the largest part of the brain. It is divided into left and right hemispheres, and can be further categorized into four main lobes, each of which is responsible for a series of specific functions.
- The surface of the cerebrum is referred to as the *cerebral cortex*. Composed of grey matter, the cortex has many folds which act to increase surface area.



The Forebrain: Cerebrum

Four Lobes of the Cerebrum

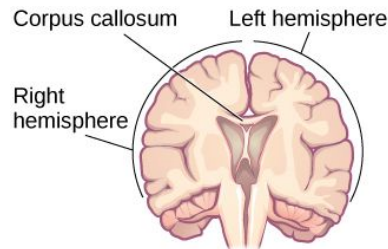
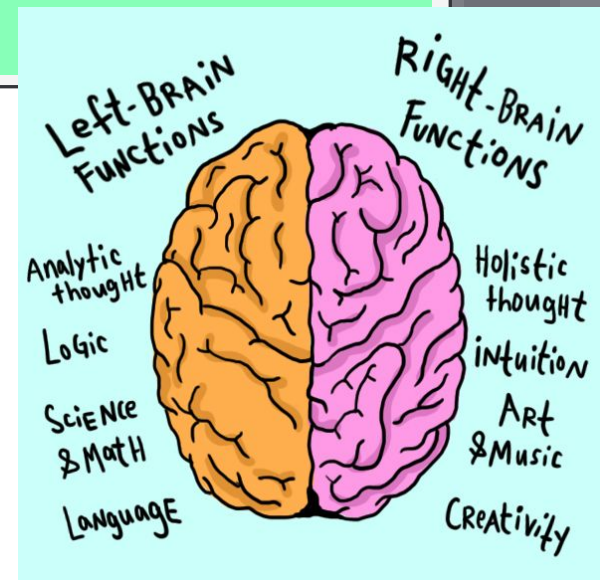
- 1) **Frontal Lobe**
 - conscious thought; decision making
 - intelligence, memory & personality
 - controls voluntary muscle movements
- 2) **Parietal Lobe**
 - somatosensory functions (touch)
 - processes information about body position
- 3) **Temporal Lobe**
 - auditory reception (hearing)
- 4) **Occipital Lobe**
 - processes visual information



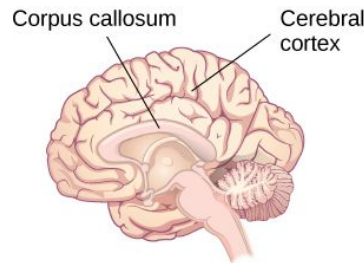
The Forebrain: Cerebrum

Left & right hemispheres

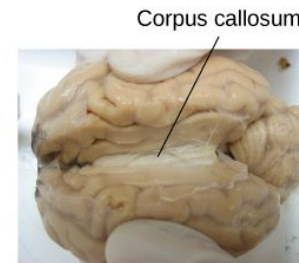
- Structures found in the left and right hemispheres are responsible for different functions
- The left-hemisphere tends to be more analytical & concrete, while the right-hemisphere tends to be more creative & abstract
 - However, there is a great degree of overlap and interaction between the
- Communication between the two hemispheres occurs via the *corpus callosum* and is essential for integrating these functions



(a) Front View



(b) Side View

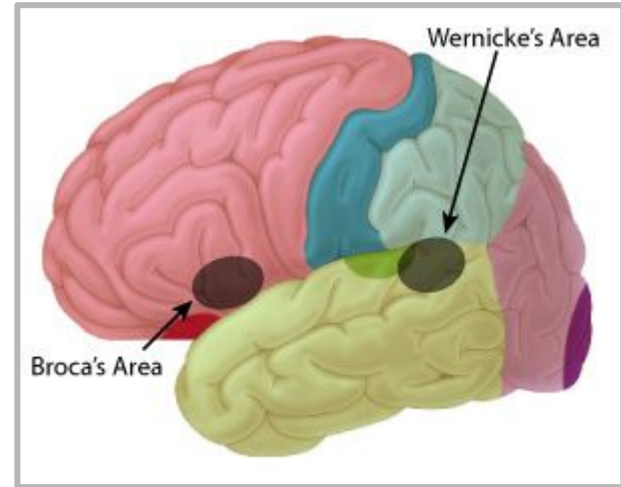


(c) Dissected Brain

The Forebrain: Cerebrum

Extra: Broca's vs. Wernicke's Aphasia

- Damage to the brain can lead to impairments in language, or *aphasia*.
- Depending on which area is damaged, patients may experience difficulties with language production (Broca's Aphasia) or language comprehension (Wernicke's Aphasia).
- In Broca's Aphasia, patients are unable to produce fluent speech despite understanding what others are saying.
- In Wernicke's Aphasia, patients are able to produce fluent speech, but do not actually understand what they are saying/what is being said to them, so their words often do not make sense.

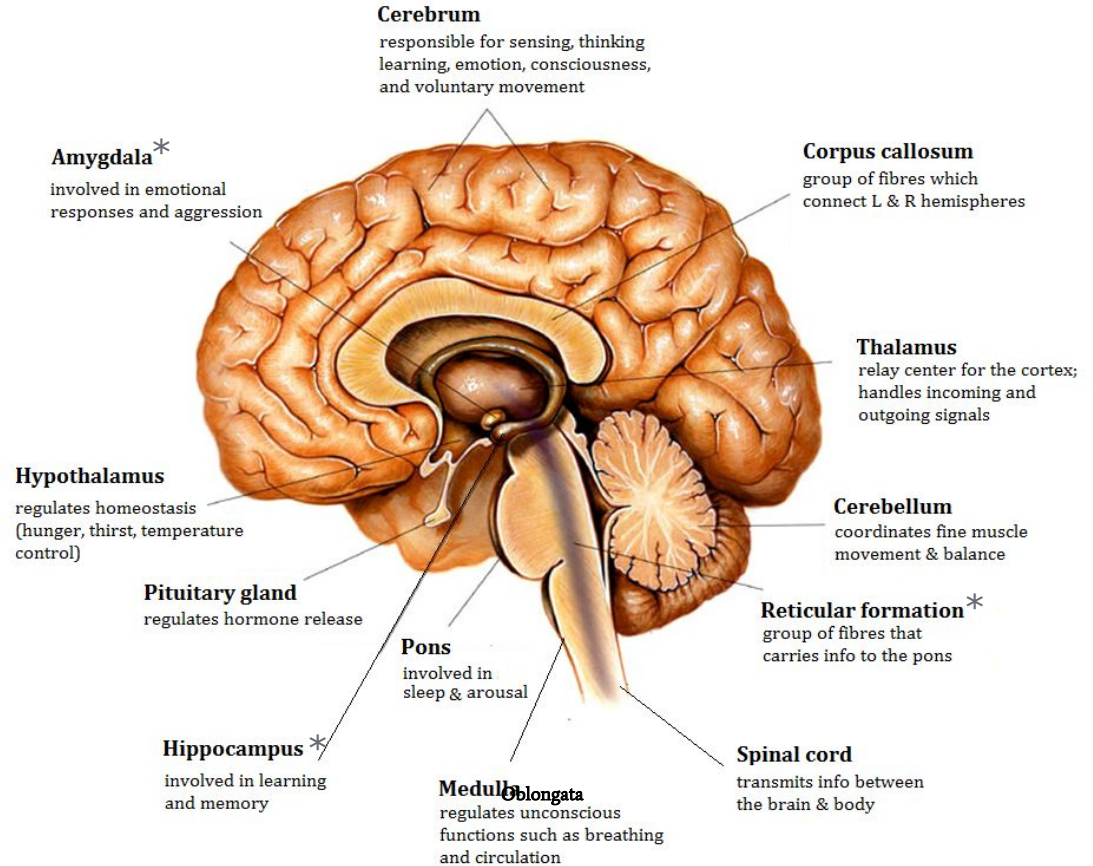


Patients with Broca's Aphasia can learn to speak through song, as the area of the brain responsible for singing is different than the area responsible for speech.

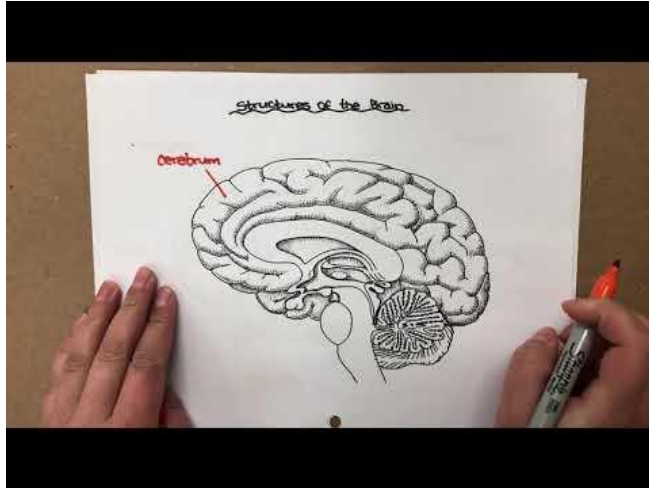
*

Other key structures of the brain & their functions:

*will not be tested



Re-cap: Structures of the Brain



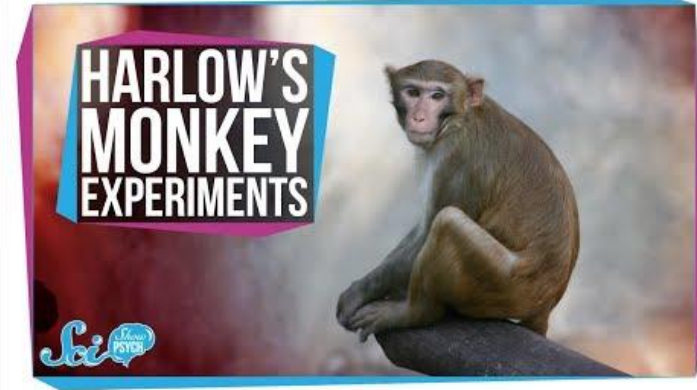
The Human Brain:
Major Structures and Functions

An in-depth understanding of the brain has
helped inform a number of misunderstood
issues...

(the next three slides are EXTRAs)

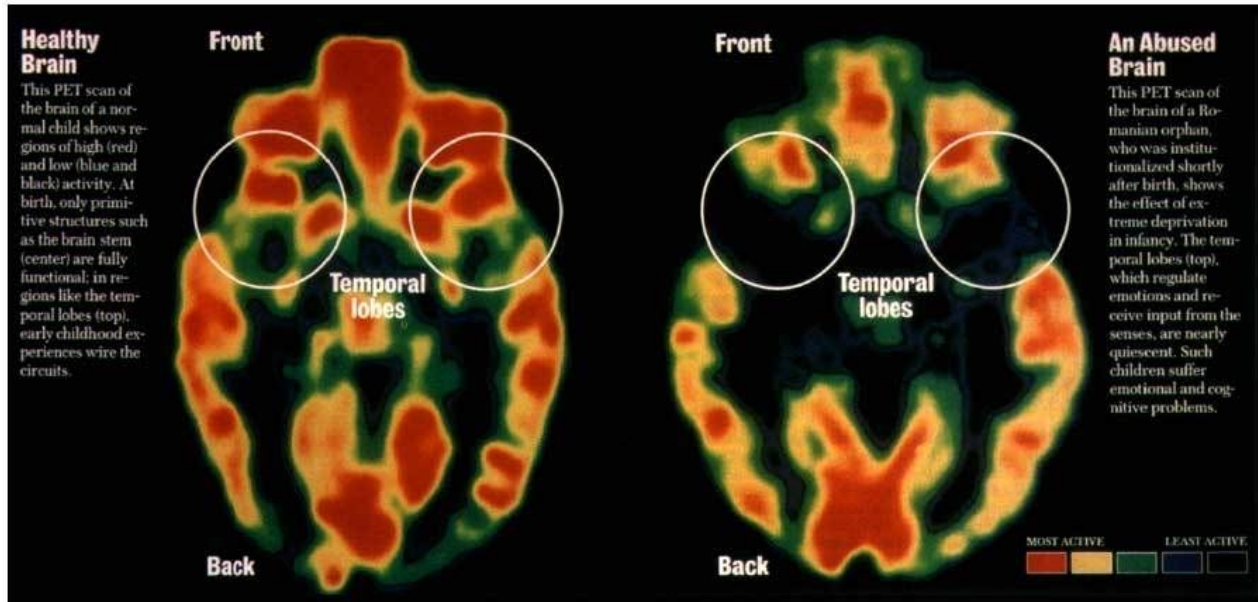


Example:
Association between
chronic
depression/anxiety
and early childhood
trauma...

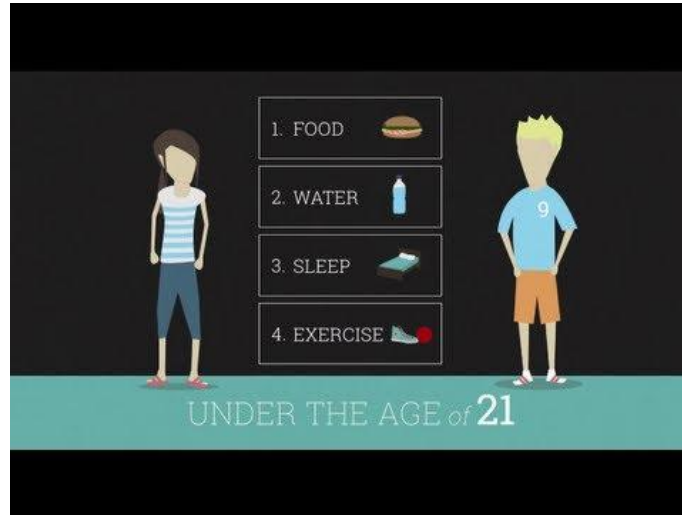


<https://www.blueknot.org.au/Resources/General-Information/Impact-on-brain>

Early childhood trauma actually changes the way the brain develops:

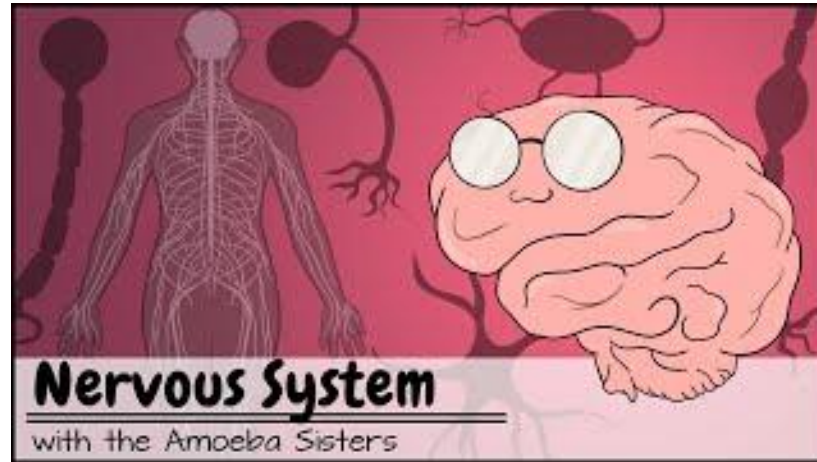


Another example: Addiction as a disease? Or a form of learning...



Re-cap: Nervous System Quick Summary

*10 minute review of everything up to this point





Chapter 12

Sensory Reception

Chapter 12: Sensory Reception

- Describe the structure and function of the parts of the human eye (e.g. cornea, lens, sclera, choroid, retina, rods & cones, fovea centralis, pupil, iris and optic nerve)
- Describe the structure and function of the parts of the human ear (e.g. pinna, auditory canal, tympanum, ossicles, cochlea, organ of Corti, auditory nerve, semicircular canals and Eustachian tube)
- Explain other ways that humans sense their environment and their spatial orientation within it (i.e. olfactory receptors, taste receptors, touch receptors)

*** Textbook pages 404-433**

Sensation

Sensation occurs when neural impulses generated by the stimulation of sensory receptors arrive at the cerebral cortex. **Perception** occurs in the brain as it makes sense of the received input.

Example:

When a ray of sunlight hits your skin, sensory receptors send a message to the brain, which receives and processes this information, causing you to feel warmth.

** each person's unique perception of sensation depends on how their cerebral cortex interprets this sensory information*



Sensory Adaptation

- Do you constantly feel your clothes against your skin? Sometimes the brain can actually filter out redundant sensory information so as to prevent over-stimulation. This is known as ***sensory adaptation*** .
- Another example is when you walk into a coffee shop and smell the fresh smell of coffee, but after a while you are not aware of it. Or when you no longer hear/notice a background noise.
- When your sensory receptors detect a significant change in external or internal conditions, the body readjusts.



Sensory Receptors

Four general categories of receptors:

1) **Photoreceptors**

Stimulated by light (e.g. rods & cones in the retina of our eyes)

2) **Chemoreceptors**

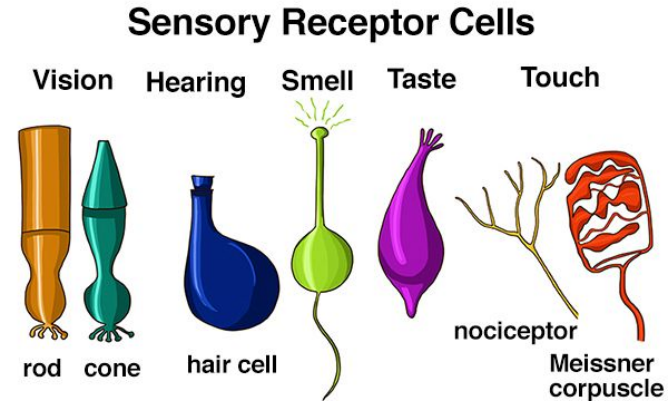
Stimulated by chemicals (e.g. taste buds on our tongue)

3) **Mechanoreceptors**

Stimulated by pressure (e.g. hair cells of the inner ear)

4) **Thermoreceptors**

Stimulated by heat/cold (e.g. nerve cells in our skin)

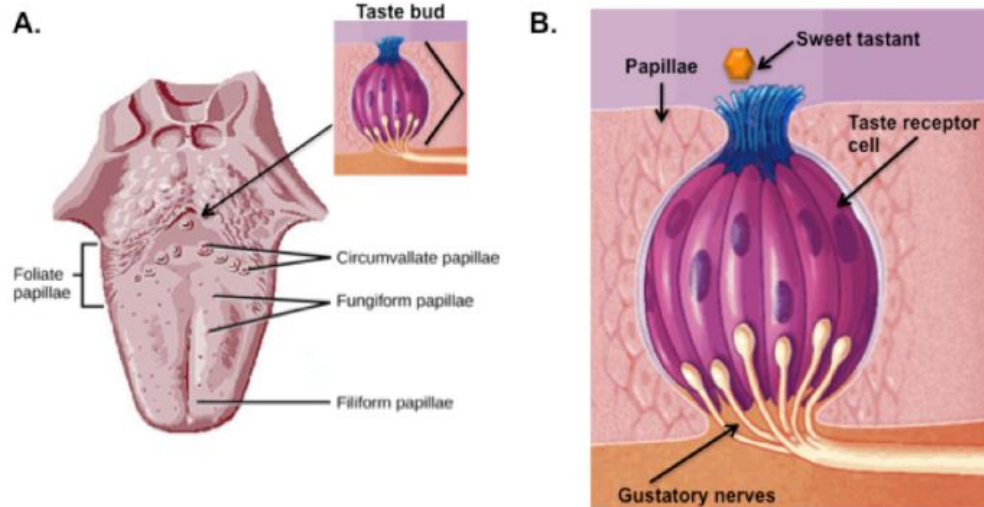


Sensory Receptor Types

Modality	Receptor	Receptor type	Organ	Stimulus
Vision	• rods and cones	• electromagnetic	• retina	• light
Hearing	• auditory hair cells	• mechanoreceptor	• organ of Corti	• vibration
Olfaction	• olfactory nerve endings	• chemoreceptor	• individual neurons	• airborne chemicals
Taste	• taste cells	• chemoreceptor	• taste bud	• food chemicals
Touch (a few examples)	• Pacinian corpuscles • free nerve endings • temperature receptors	• mechanoreceptor • nociceptor • thermoreceptor	• skin	• pressure • pain • temperature

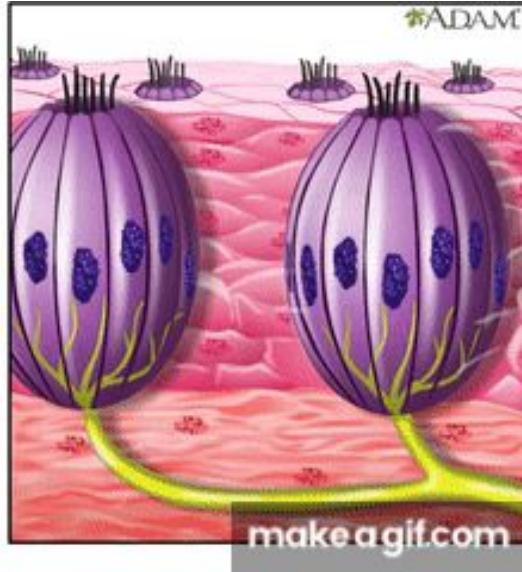
Taste (Chemoreception)

The tongue contains chemoreceptors that allow us to taste substances entering the mouth. Impulses from **taste buds**, where chemoreceptors are located, then travel to areas of the brain responsible for the perception of taste (parietal lobe)



Taste (Chemoreception)

Food particles are detected by the taste buds which relay a signal to his brain.



Extra: Five major types of taste receptors dispersed throughout the tongue:

1. sour
2. sweet
3. salty
4. bitter
5. unami

Olfaction/Smell (Chemoreception)

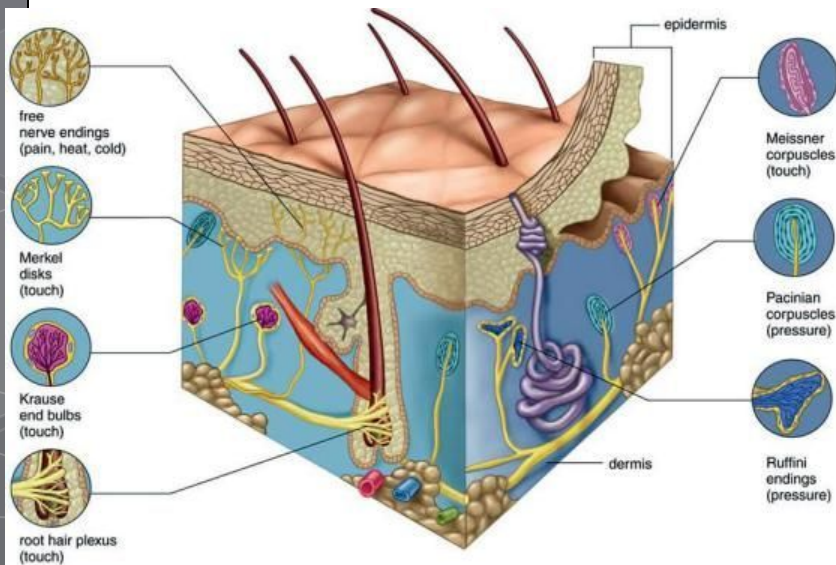
Humans can distinguish between over 10,000 different odours. Each of these odour particles is thought to fit like a lock & key into specific chemoreceptors which line the nasal cavity, known as ***olfactory cells*** .

- When stimulated, olfactory cells transmit signals to the olfactory bulb in the cerebrum of the brain, leading to the sensation of smell
- The olfactory bulb of dogs is more than twice the size of humans, explaining their heightened sense of smell



Receptors in the Skin

- There are many different types of receptors in the skin that respond to a variety of different stimuli such as touch, pressure, temperature, pain, and various intensities of these. (You do not need to have any of this memorized, just appreciate the complexity!)



RECEPTOR TYPE	RECEPTOR STRUCTURE AND LOCATION	SENSATIONS
TACTILE RECEPTORS		
Corpuscles of touch (Meissner corpuscles)	Capsule surrounds mass of dendrites in dermal papillae of hairless skin.	Touch, pressure, and slow vibrations.
Hair root plexuses	Free nerve endings wrapped around hair follicles in skin.	Touch.
Type I cutaneous mechanoreceptors (tactile discs)	Saucer-shaped free nerve endings make contact with tactile epithelial cells in epidermis.	Touch and pressure.
Type II cutaneous mechanoreceptors (Ruffini corpuscles)	Elongated capsule surrounds dendrites deep in dermis and in ligaments and tendons.	Touch and stretching of skin.
Lamellated (Pacinian) corpuscles	Oval, layered capsule surrounds dendrites; present in dermis and subcutaneous layer, submucosal tissues, joints, periosteum, and some viscera.	Pressure and fast vibrations.
Itch and tickle receptors	Free nerve endings in skin and mucous membranes.	Itching and tickling.
THERMORECEPTORS		
Warm receptors and cold receptors	Free nerve endings in skin and mucous membranes of mouth, vagina, and anus.	Warmth or cold.
PAIN RECEPTORS		
Nociceptors	Free nerve endings in every body tissue except brain.	Pain.
PROPRIOCEPTORS		
Muscle spindles	Sensory nerve endings wrap around central area of encapsulated intrafusal muscle fibers within most skeletal muscles.	Muscle length.
Tendon organs	Capsule encloses collagen fibers and sensory nerve endings at junction of tendon and muscle.	Muscle tension.

Receptors in the Skin

- We should, however, appreciate that the distribution of many receptors, such as touch receptors, are not uniform throughout the body and skin
- For example, you have a **much greater density of touch receptors in your fingertips or your lips and tongue**, than on your ankle or your back.
 - This is why we are so bad at the drawing on the back game!



Vision (Photoreception)

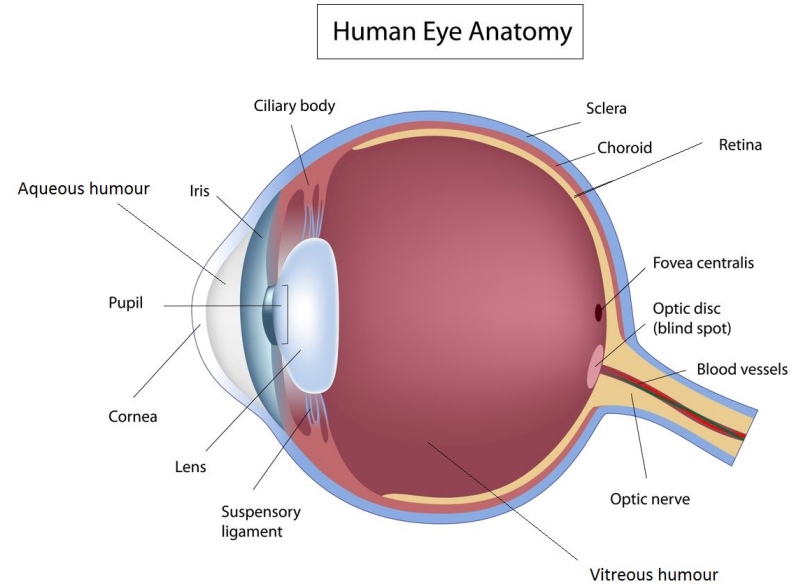
Vision is associated with the stimulation of **photoreceptors** at the back of the eye by light energy.

Rods (light sensors → black/white vision) and **cones** (colour sensors) in the retina convert this light energy into electrochemical impulses which the brain uses to form an image...



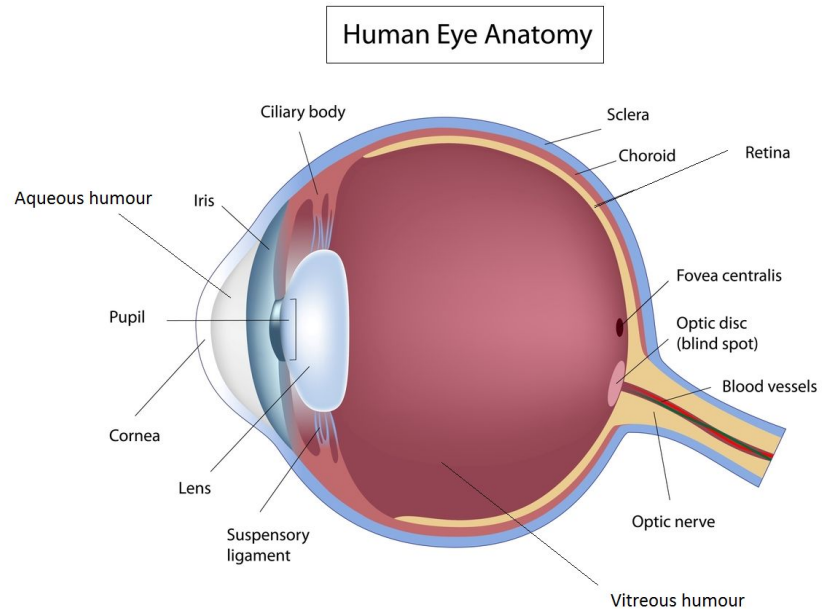
Anatomy of the Eye & Vision

- Light enters the eye through the **cornea**, which is the clear part of the **sclera** in front of the eye.
- The **iris**, is the colored part of the eye and is a muscle that contracts and relaxes to control how much light actually enters the eye through the **pupil**, the hole in the iris.
- **Ciliary muscles** hold the **lens** in place and control its shape, allowing it to focus on objects closer or further away and focus the incoming light onto the **retina**. (this process is called *accommodation*)
- The focused light strikes the **retina**, which is the inner layer of the back of the eye. The retina is composed of **rods** (sensitive to light intensity) and **cones** (sensitive to wavelengths of colour)
- Photoreceptors (rods and cones) relay sensory impulses to the occipital lobe of the brain via the **optic nerve**

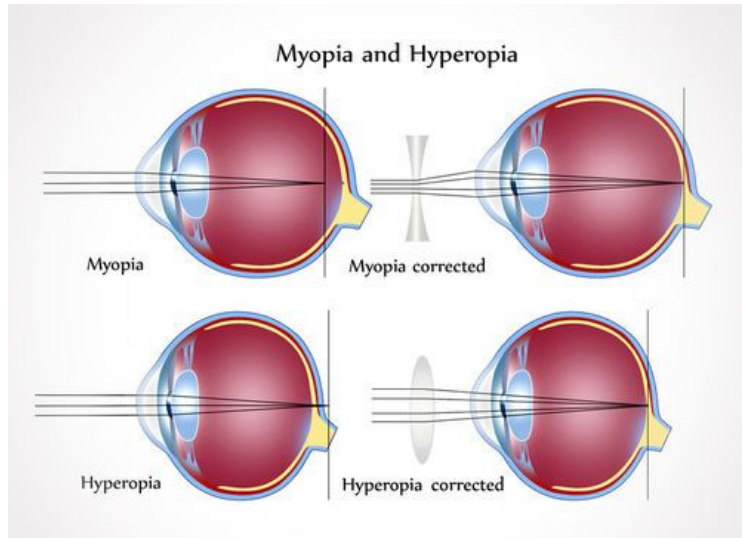


Anatomy of the Eye & Vision

- The **aqueous humour** supplies the cornea with nutrients, while the **vitreous humour** maintains the shape of the eye and permits light transmission to the retina.
- The **sclera** protects the eye and helps to maintain its shape.
- The **choroid layer** contains blood vessels which nourish the retina.
- The **fovea centralis** is a location in the retina that contains a high concentration of cones, while the **blind spot** (the point at which the optic nerve connects to the retina) contains neither rods nor cones. This is why when light falls on the blind spot, no image is generated.



Extra: Conditions Affecting the Shape of the Eye



- **Myopia (nearsightedness)**
 - the inability to see objects at a distance
 - eyeball is slightly elongated, so focused light falls in front of the retina
 - nearsighted people can wear concave lenses to help focus light directly on the retina

- **Hyperopia (farsightedness)**
 - the inability to see objects at close range
 - eyeball is slightly shortened, so focused light falls behind the retina
 - farsighted people can wear convex lenses to help focus light directly on the retina

Rods vs. Cones

- The retina is composed of two types of light-sensitive photoreceptors: **rods and cones**.
- Rods respond to low-intensity light to detect black and white.
 - We rely on rods during low light or nighttime vision
- Cones respond to high-intensity light to identify colour.
 - We rely on cones to general high visual detail when there is an abundance of light

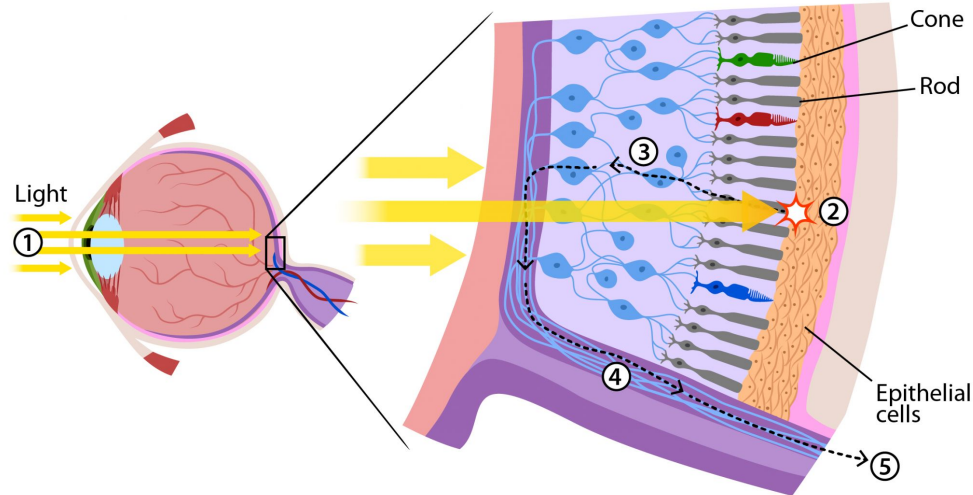
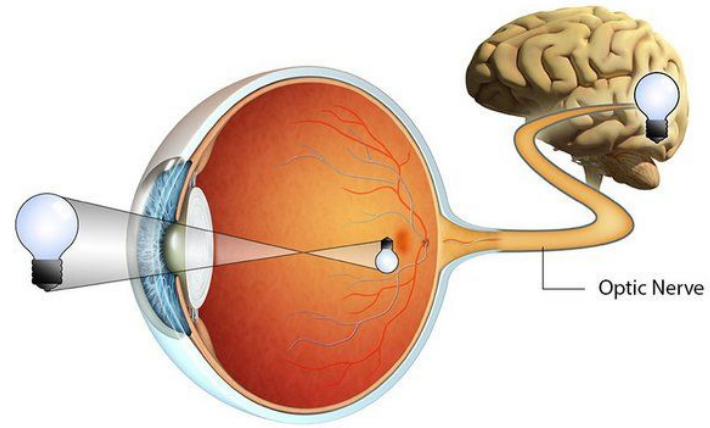


Image Generation

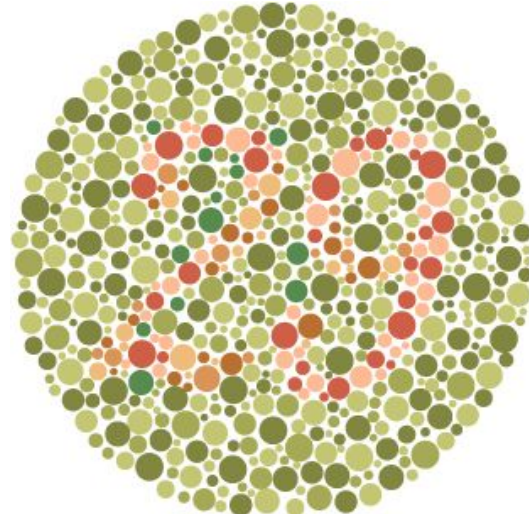
- The lens generates an inverted image on the retina, but the brain flips the received message to generate an upright perceived image.



Conditions Affecting the Cones

Colorblindness is a genetic condition caused by a lack of specific cones, or colour receptors.

- Most color blind individuals lack the cones which respond to red-green wavelengths, so they cannot perceive those colours (see them as shades of brown)



what do you see???

Conditions Affecting the Rods

- Rods contain a light-sensitive pigment called *rhodopsin*, a form of vitamin A. Thus, a lack of vitamin A can lead to what is commonly referred to as “night blindness”.



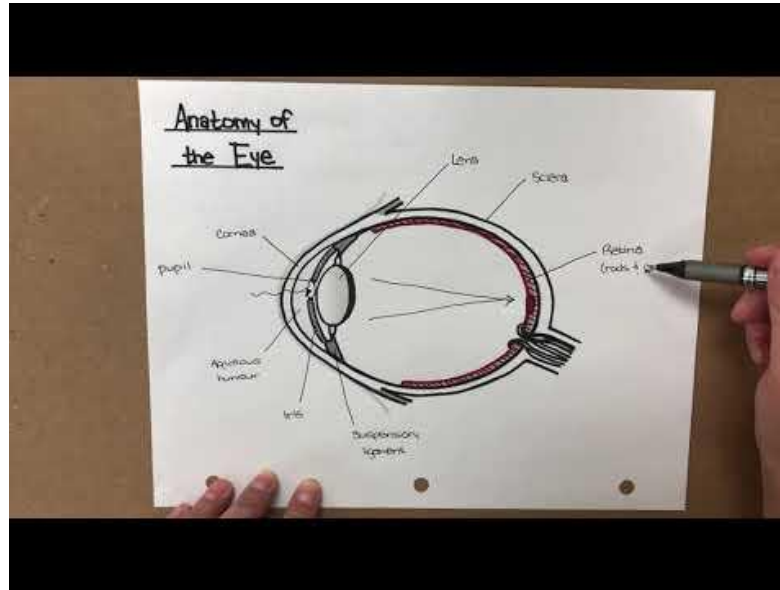
Extra/Fun Fact: Vision Development

In humans, development of the eye and brain structures is only partially complete upon birth.

- Newborns are thought to see the world “upside down” for the first few days until their brain learns to properly interpret the info it is receiving



Re-cap: Anatomy of the Eye



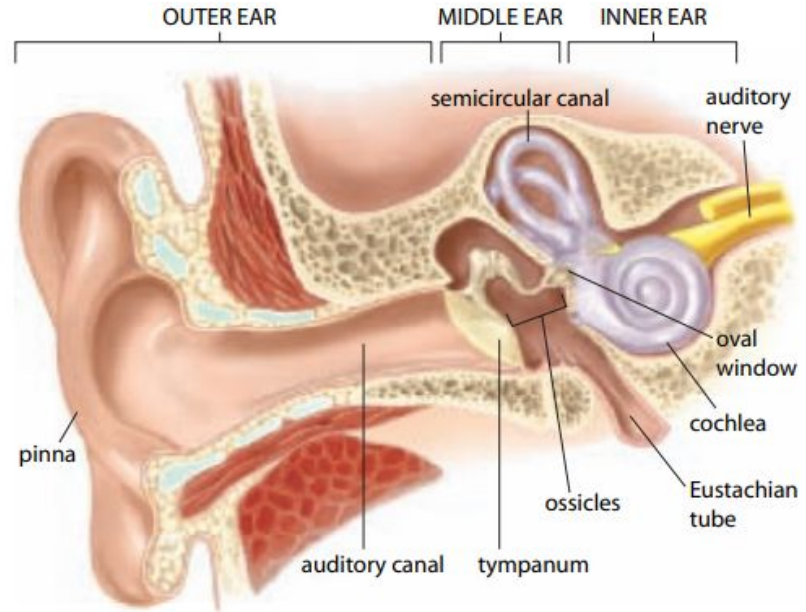
Hearing (Mechanoreception)

Hearing is associated with small fluctuations in air pressure, called *sound waves*. Mechanoreceptors in the inner ear convert the energy of sound waves into electrochemical impulses which the brain perceives as sound...



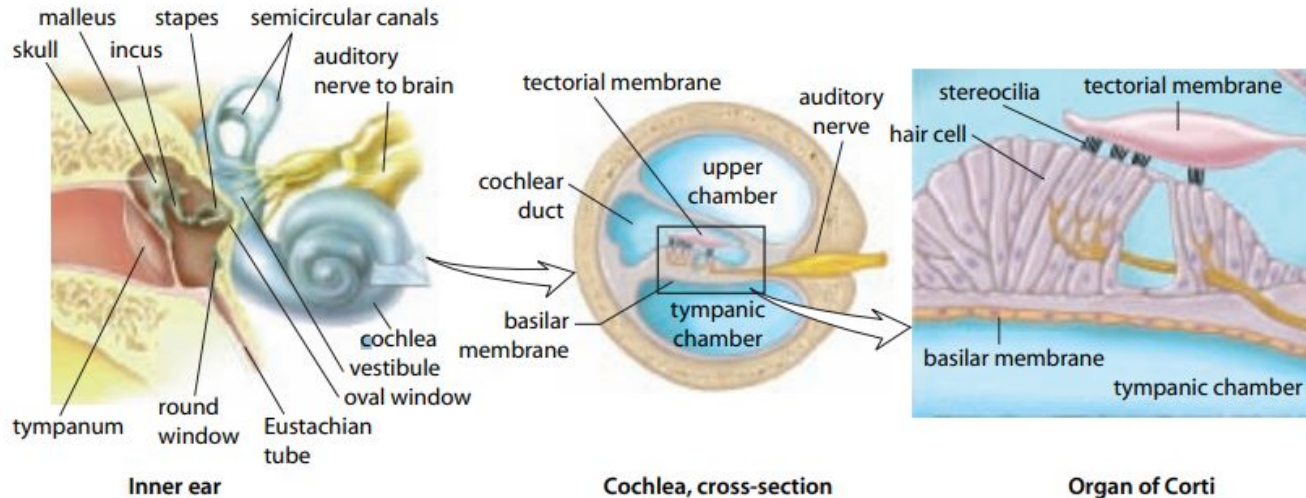
Anatomy of the Outer/Middle Ear

- Sound waves are directed by **pinna (auricle)** to the **auditory canal**, where they strike the **tympanum** (eardrum), generating vibrations within the middle ear.
- When sound waves push against the tympanum, its vibrations are passed on and amplified by a set of three small connected bones, called the **ossicles**
- Vibrations are concentrated on the **oval window**, causing changes in pressure within the organs of the inner ear

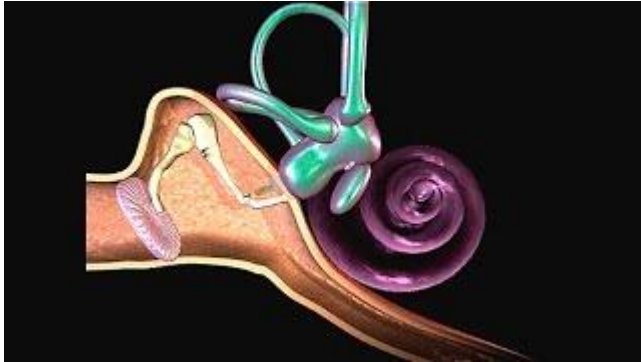


Anatomy of the Inner Ear

- Changes in pressure stimulate the movement of hair cells (collectively known as the **organ of Corti**) within the coiled **cochlea** , generating an action potential within the **auditory nerve**.
- This action potential is carried to the temporal lobes by the auditory nerve, where sound information is processed and interpreted.

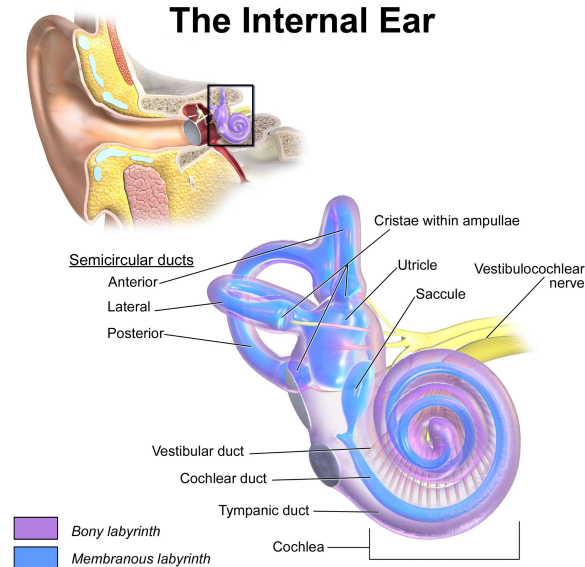
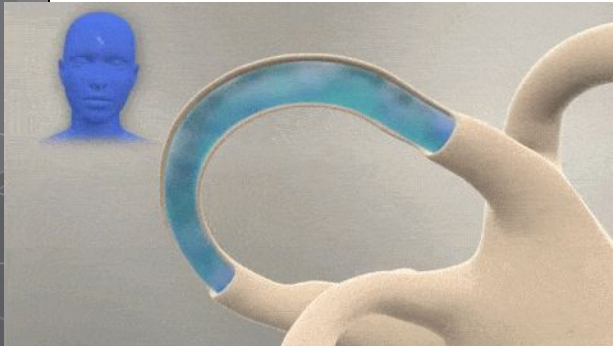


Hearing and Sound Perception



Parts of the Ear not Involved with Hearing

- The **vestibular system** acts to provide a sense of **balance** and information about **body positioning**
- Two main components:
 - **Semicircular canals** : Three fluid filled canals in each of the three dimensions with hair cells that respond to movement → provides information about head rotation and tilting
 - *Extra: Utricle and Saccule: Provide information about body position and acceleration*



Extra: Balance Disorders

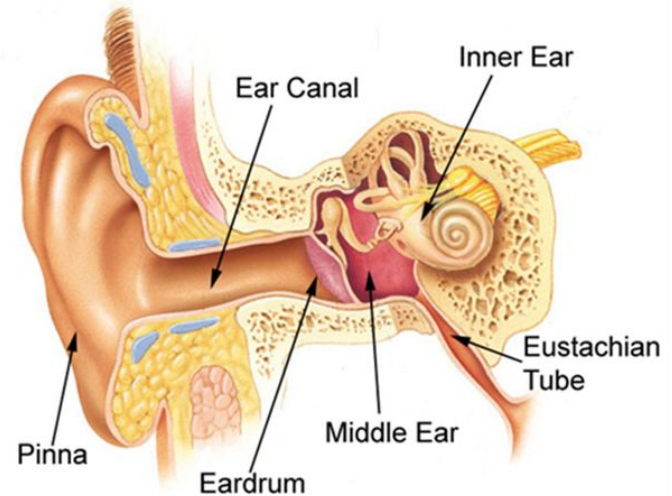
Balance disorders commonly result from damage to the semicircular canals, including..

- leakage of inner ear fluid after head injury or dramatic changes in air pressure, causing dizziness and nausea
- failure of canals to flex properly in response to a change in head position, causing dizziness (vertigo)
- an increase in fluid volume, leading to a ringing or buzzing and a feeling of fullness in the ear



Parts of the Ear not Involved with Hearing

- The **Eustachian tube** is an opening that connects the middle ear with the nasal cavity, and acts to equalize pressure in the middle ear. This is what is commonly felt as your ears popping



Extra: Hearing Loss

Hearing loss may be caused by a number of factors, including:

- **Birth defects**

Improper development of any of the organs required for hearing can lead to deafness; often hereditary, but can sometimes be treated with cochlear implants

- **Ear infections**

Swelling of the middle ear can prevent sound waves from reaching the cochlea; in some cases, structures of the middle ear can become permanently damaged

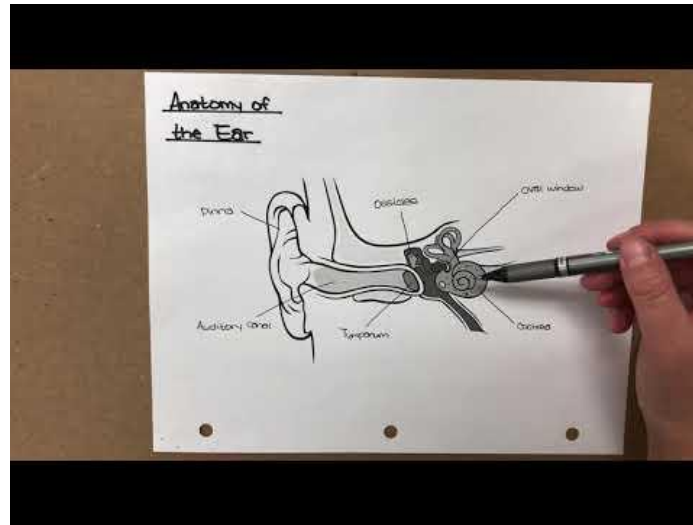
- **Noise**

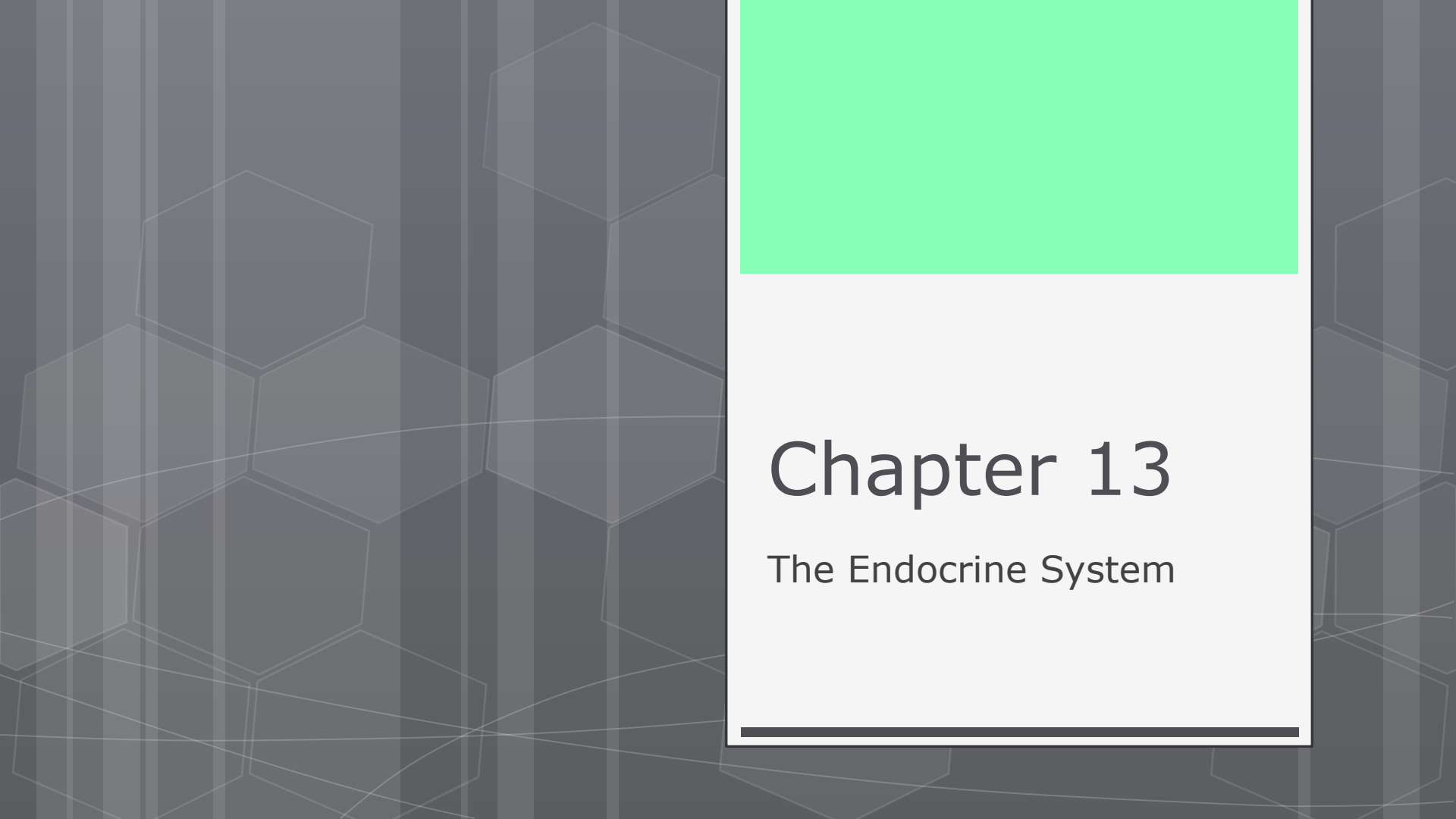
Loud noises can damage hair cells within the cochlea or even cause the tympanic membrane to rupture



<https://bottomlineinc.com/health/hearing-aids/soundbite-hearing-aid-worn-inside-the-mouth>

Re-cap: Anatomy of the Ear





Chapter 13

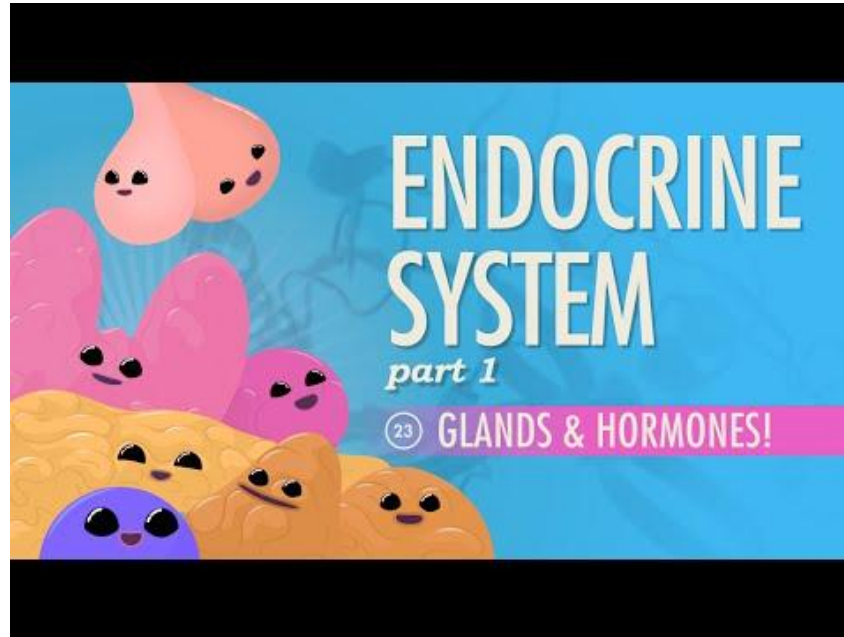
The Endocrine System

Chapter 13: Endocrine System

- Roles of the endocrine vs. nervous system
- Positive vs. negative feedback
- Principal endocrine glands: hypothalamus/pituitary complex, thyroid, parathyroid, adrenal glands, & islet cells of the pancreas

*** Textbook pages 434-465**

Intro/Overview

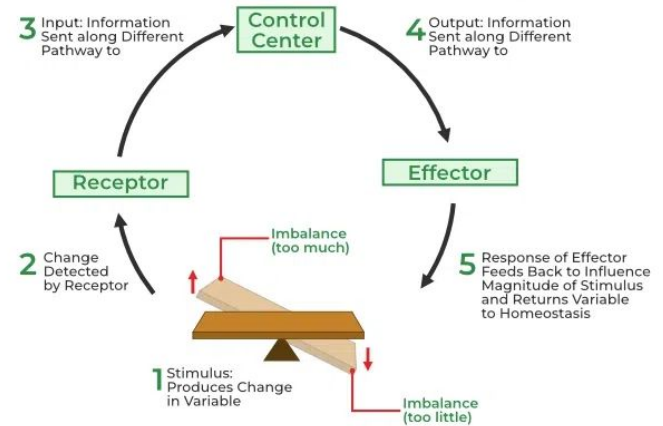
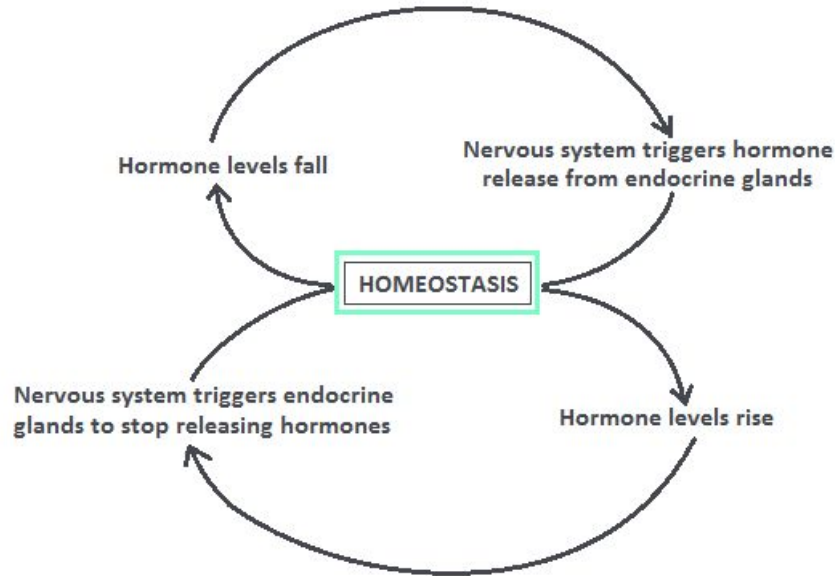


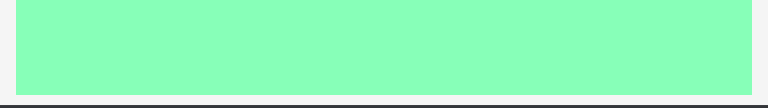
Homeostasis is...

- A stable internal environment that the body aims to maintain for optimal functioning
- A 'state of internal equilibrium'



The endocrine system works in conjunction with the nervous system to maintain homeostasis within the body:



- 
- Although the two systems work together, there are some key differences between them and the way in which they act

NERVOUS SYSTEM	ENDOCRINE SYSTEM
Rapid response to change	Sustained response to change
Electrical/chemical signals	Hormonal signals
Signals relayed through neurons (hence rapid response)	Signals relayed through the bloodstream (hence slower response)
Affects specific locations	Affects whole-body functioning

Feedback Loops

The endocrine system responds to changes in the internal environment through the use of positive or negative feedback.

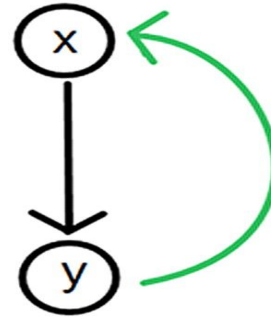
Negative Feedback:

- returns a system to homeostasis
- inhibits or counteracts the stimulus that sets off the response

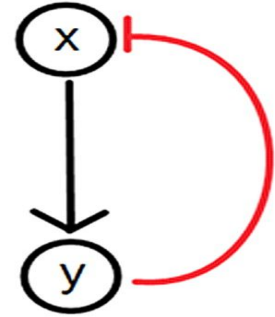
Positive Feedback:

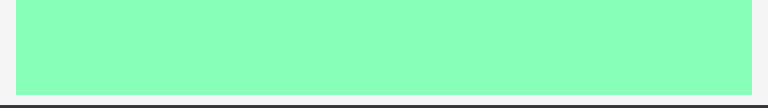
- takes the body further from homeostasis
- exaggerates or increases the stimulus that sets off the response

Positive



Negative





Feedback Loop Examples

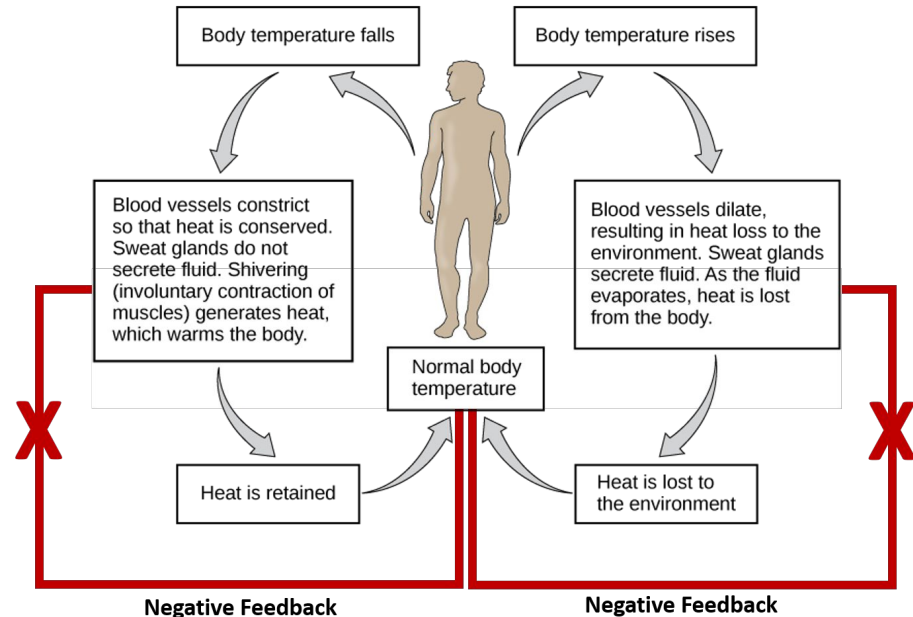
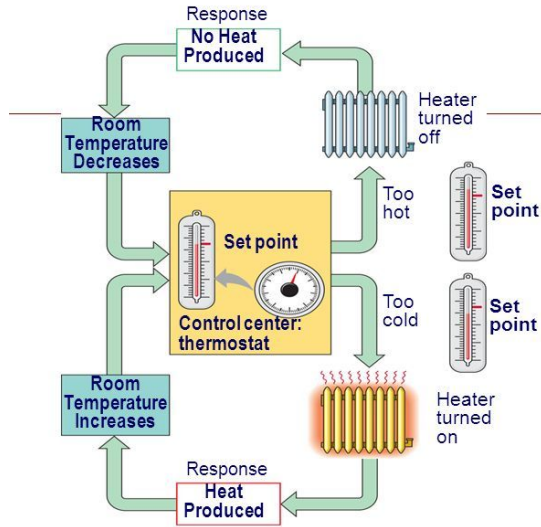
Negative Feedback:

- Blood sugar: After eating, blood sugar levels rise, triggering the release of insulin from the pancreas. Insulin stimulates cells to become permeable to sugar, causing blood sugar levels to fall, and insulin production to cease.)

Positive Feedback:

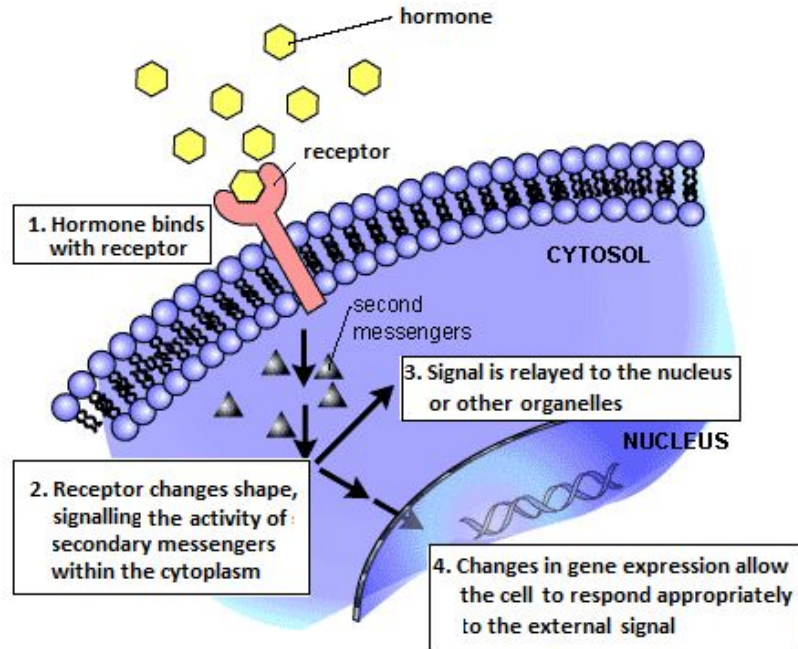
- Childbirth: As the baby moves into position and presses on the cervix, it sends a signal to increase the rate of contractions, which further pushes the baby against the cervix which in turn increases contractions even further.

- Most feedback loops in the human body are negative, and aim to reestablish homeostasis when a particular value is out of normal range.
- An example is body temperature, that can be compared to how the thermostat works in your home:



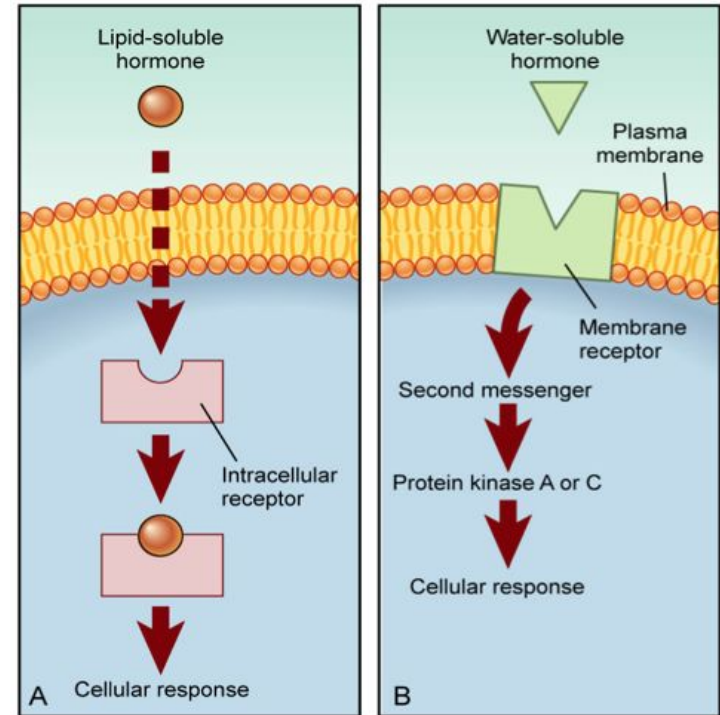
Hormones

- Hormones interact with receptor proteins specific to their target tissue, inducing a conformational shape change which leads to a cascade of events within each cell



Two types of hormones, classified by the way they interact with target cells:

- Water soluble hormones cannot cross the lipid membrane, and thus interact with cells by binding to external receptor sites (e.g. hGH, epinephrine)
- Fat soluble hormones are able to diffuse across the lipid membrane, and thus interact with cells by binding to receptor sites within the cytoplasm (e.g. sex hormones)



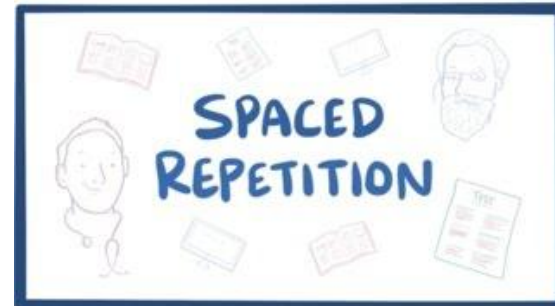
Over the next several slides, you will learn about each endocrine gland and its hormones. There is a LOT to remember for this chapter, which can include some memorization. **BUT**..memorization should not occur without meaning. I strongly encourage you to watch these two short videos that will help you use some strategies to better retain and recall information, not only in Bio but in your other courses as well.

Memorization Methods

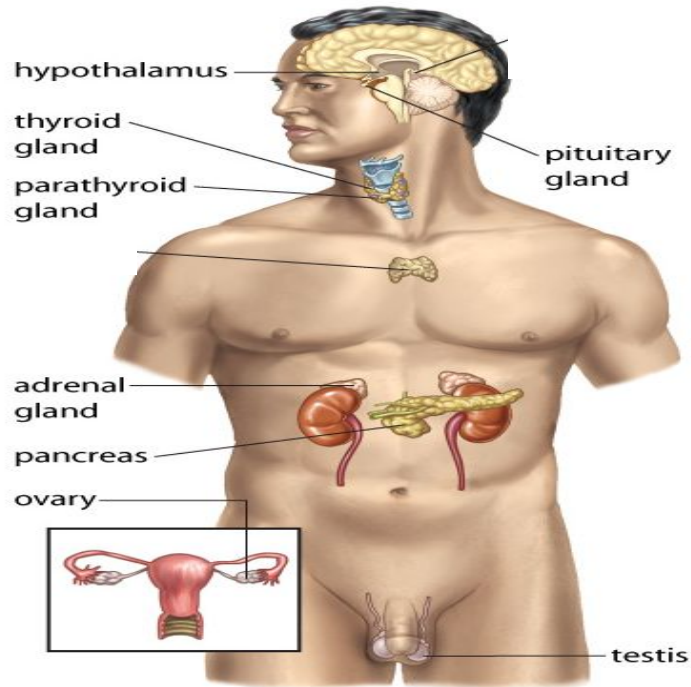


You are responsible for knowing:

1. ALL of these hormones
2. The gland that releases them
3. Their target
4. Their effect



Key glands of the endocrine system

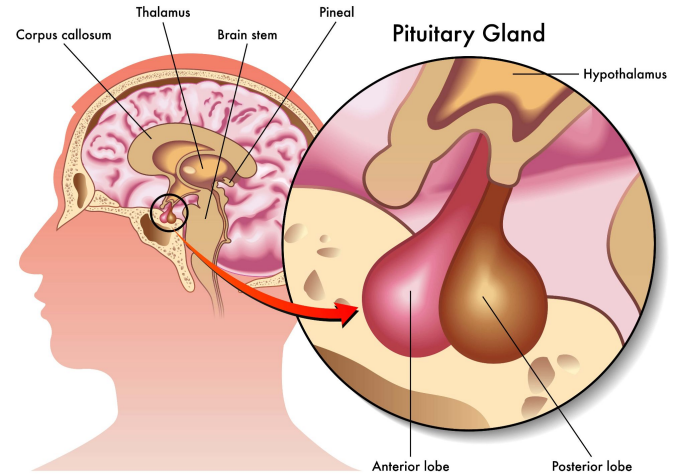


Each is responsible for secreting specific hormones which, in turn, target and affect the function of specific tissues/organs.

(see table 13.1 in your text)

Hypothalamus & Pituitary

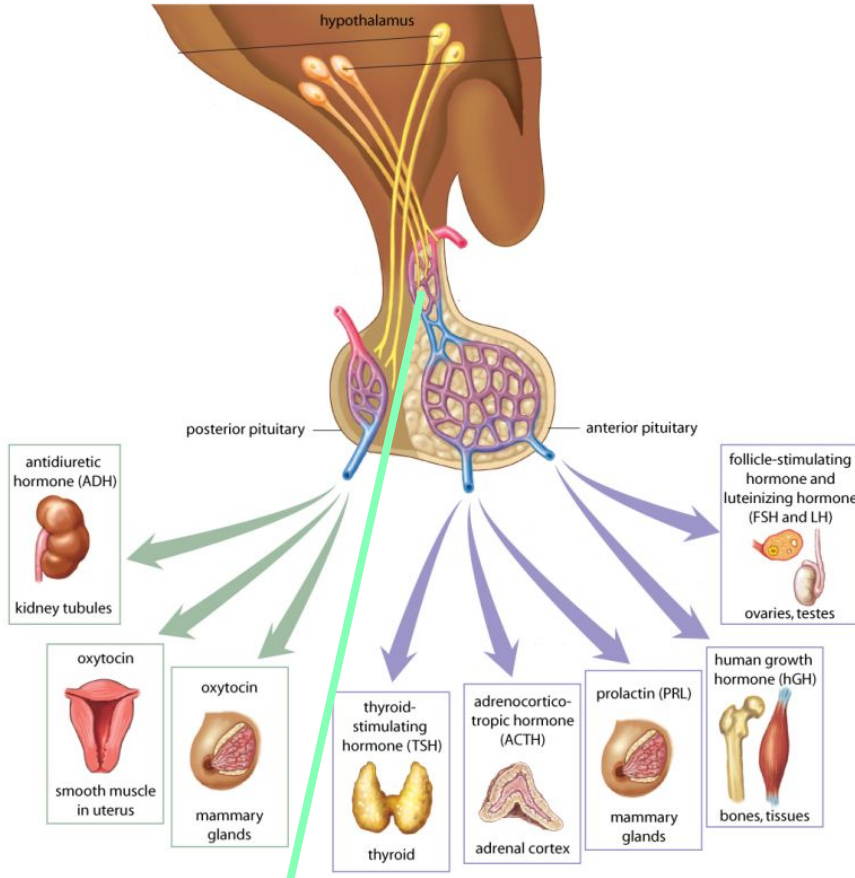
- The **hypothalamus** uses its **releasing hormones (RHs)** to stimulate the **pituitary gland**, which in turn is responsible for releasing a number of different hormones involved in growth & metabolism, development, reproduction, and other critical functions.
- For this reason, the pituitary is often called the "*master gland*"
- Many hormones released from hypothalamus & pituitary are **tropic hormones**, meaning they stimulate other endocrine glands to release other hormones.



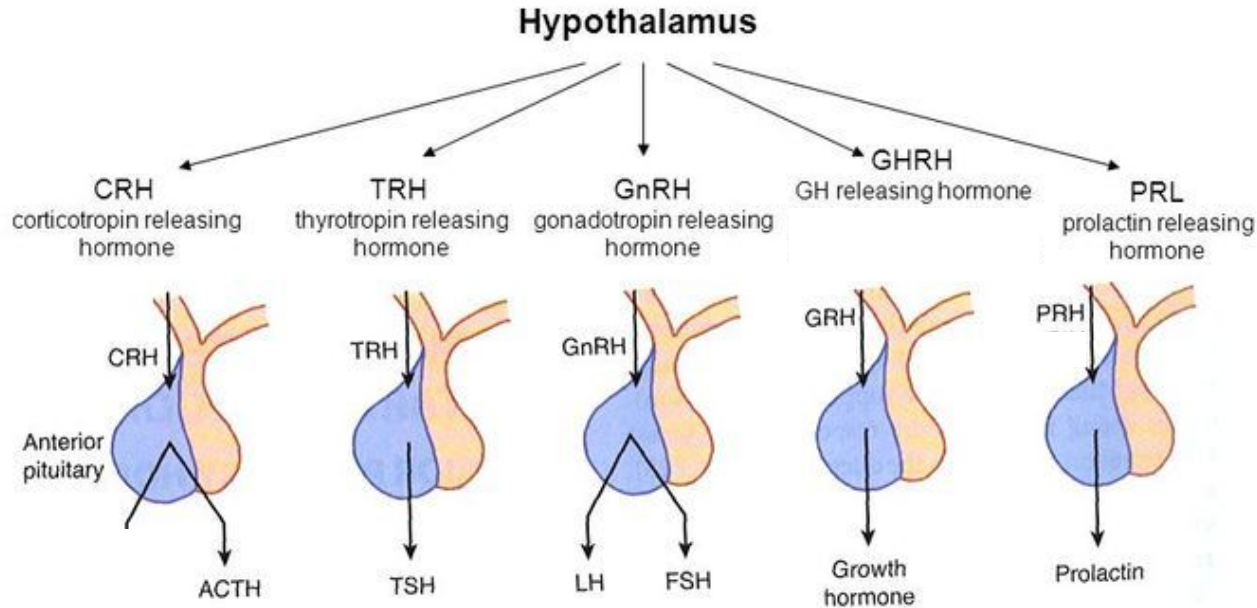
Hypothalamus & Pituitary

○ The **anterior pituitary** (front portion) synthesizes its own hormones to be released upon stimulation by hypothalamic releasing hormones

○ The **posterior pituitary** (back portion) stores hormones that were produced by the hypothalamus until release



Each anterior pituitary hormone is released upon stimulation by a 'releasing hormone' from the hypothalamus. (You only need to remember/recognize the starred ones)



Anterior Pituitary Hormones

Human growth hormone (hGH)

- affects nearly all body tissues in some way by stimulating the production of growth factors from the liver
- increases protein synthesis, cell division and growth (of cartilage, bone, and muscle tissue in particular), and metabolic release of stored fats
- Especially important during childhood, leading to the elongation of bones

Pituitary Dwarfism



Gigantism

Anterior Pituitary Hormones

Prolactin (PRO)

- Prolactin stimulates the production of milk (**lactation**) in the mammary glands

Follicle Stimulating Hormone (FSH)

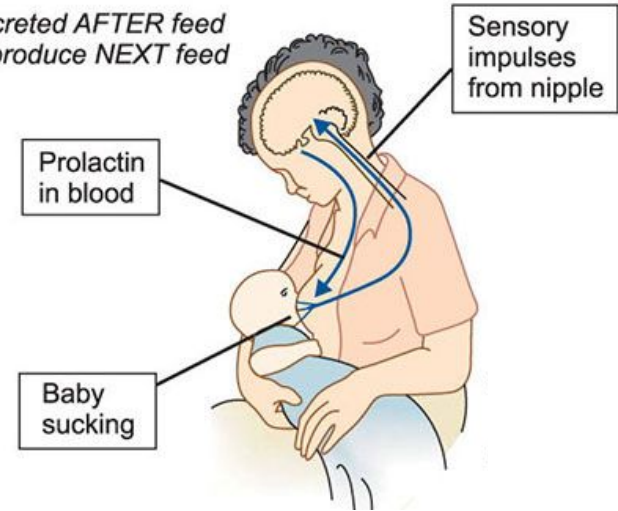
- Stimulates the production of sperm in the testes and egg in the ovaries (more on this in Unit B)

Luteinizing Hormone (LH)

- Stimulates sex hormone production in the ovaries and testes
- Stimulates ovulation (more on this in Unit B)

PROLACTIN

*Secreted AFTER feed
to produce NEXT feed*



Anterior Pituitary Hormones

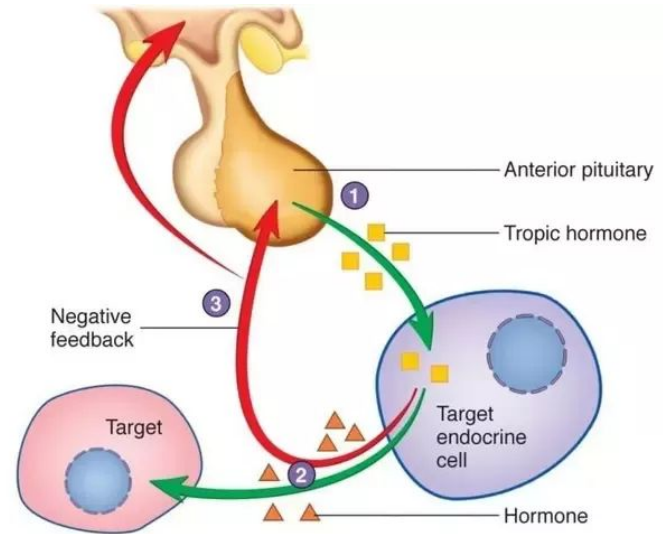
These last two anterior pituitary hormones are **tropic** hormones, which means they act on another endocrine gland. We will look at these in more detail as we look at their target glands.

Thyroid Stimulating Hormone (TSH)

- Stimulates the **thyroid gland** to release thyroxine when thyroxine levels are low

Adrenocorticotrophic Hormone (ACTH)

- Stimulates the **adrenal cortex** to release cortisol and aldosterone



Posterior Pituitary Hormones

Antidiuretic hormone (ADH)

- o also known as vasopressin
- o Released when dehydrated and regulates water reabsorption by kidneys
- o Making the collecting ducts of the nephrons more permeable to water and thus increases the reabsorption of water into the blood → this in turn makes less, more concentrated urine

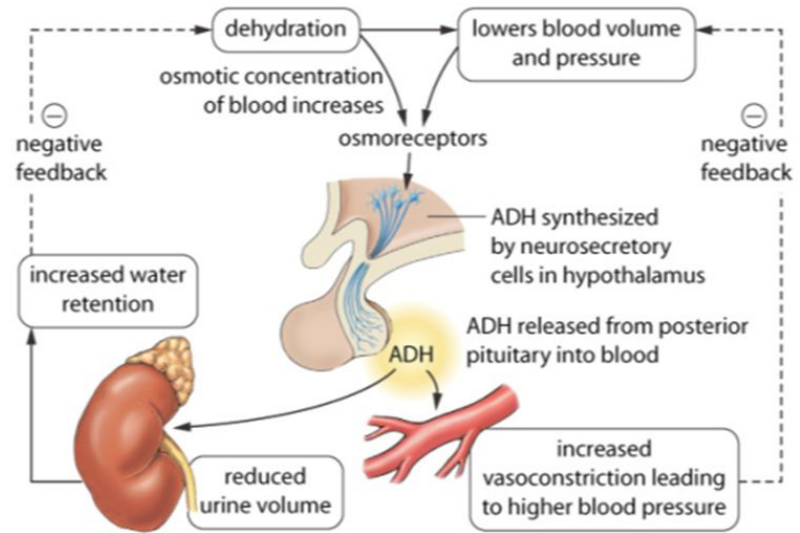


Figure 13.9 A negative feedback mechanism controls the secretion of ADH from the hypothalamus.

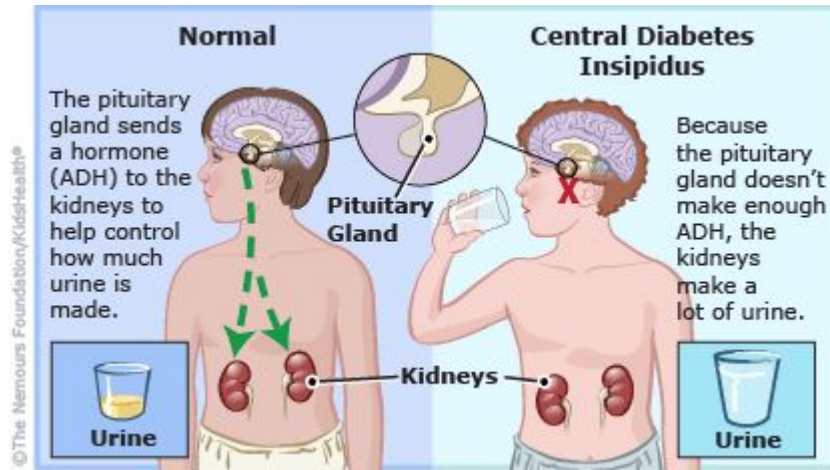


"With hindsight, maybe I shouldn't have prescribed diuretics."

- **Diuretics**, such as caffeine, inhibit the production of ADH, which in turn makes the kidneys less permeable water, leading to decreased water reabsorption.
- This leads to the production of large volumes of dilute urine.

Diabetes Insipidus

The inability to produce, store, or properly respond to ADH



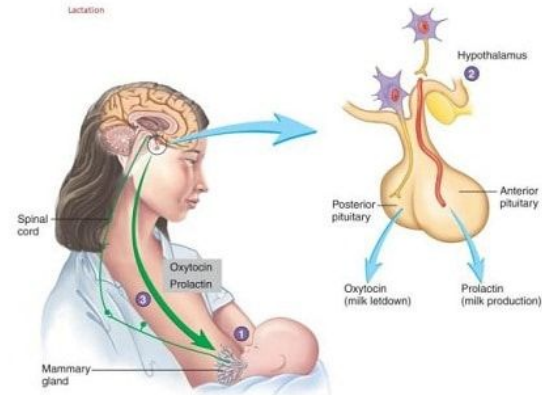
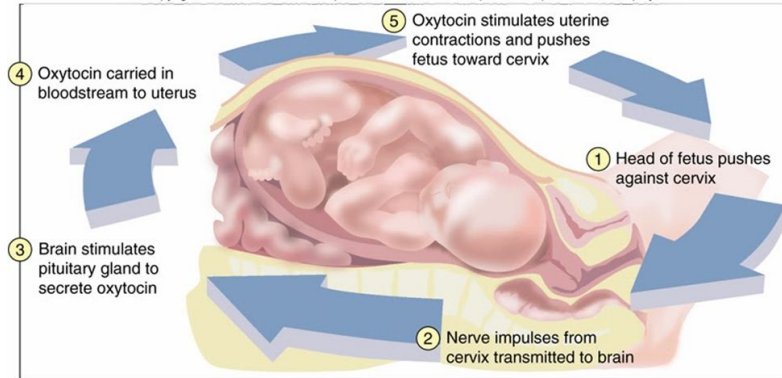
- Affected individuals produce large volumes of urine, resulting in the loss of ions from the blood
- Continuous risk of dehydration
- Treated by administering ADH

**Note: Diabetes Insipidus has nothing to do with Diabetes Mellitus (more later). They simply share the name due to similarities in symptoms.*

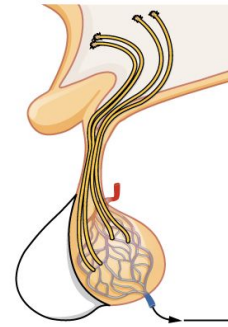
Posterior Pituitary Hormones

Oxytocin (OCT)

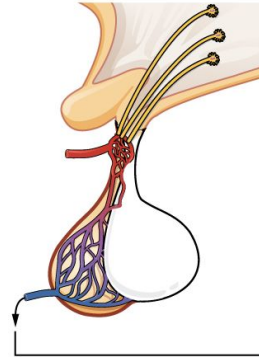
- Stimulates uterine contractions during childbirth in a positive feedback loop
- Releases milk from the mammary glands
 - Note: Prolactin allows mammals to produce milk, but oxytocin is required to release to milk
- In other words, oxytocin is the hormone that helps ‘squeeze things out of the body’



Summary of Hypothalamic and Pituitary Hormones



Posterior Pituitary Hormones				
Releasing hormone (hypothalamus)	Pituitary hormone	Target	Effects	
ADH	Stores ADH	Kidneys, sweat glands, circulatory system	Water balance	
-	OT	Female reproductive system	Triggers uterine contractions during childbirth	

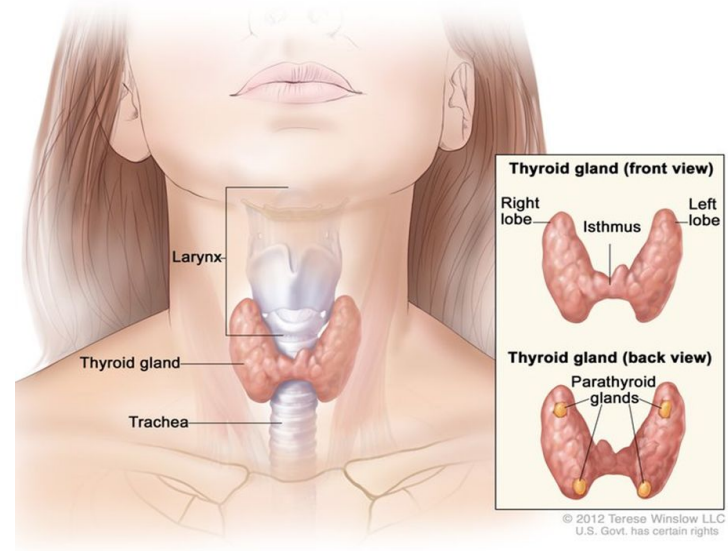


Anterior Pituitary Hormones				
Releasing hormone (hypothalamus)	Pituitary hormone	Target	Effects	
GnRH	LH	Reproductive system	Stimulates production of sex hormones by gonads	
GnRH	FSH	Reproductive system	Stimulates production of sperm and eggs	
TRH	TSH	Thyroid gland	Stimulates the release of thyroid hormone (TH). TH regulates metabolism.	
PRH	PRL	Mammary glands	Promotes milk production	
GHRH	GH	Liver, bone, muscles	Induces targets to produce insulin-like growth factors (IGF). IGFs stimulate body growth and a higher metabolic rate.	
CRH	ACTH	Adrenal glands	Induces targets to produce glucocorticoids, which regulate metabolism and the stress response	

Hormones of the Thyroid and Parathyroid Glands

The **thyroid gland** is a butterfly shaped gland that sits anterior to the trachea and below the larynx. On the back side of the thyroid gland, there are four smaller glands called the **parathyroid glands**.

Anatomy of the Thyroid and Parathyroid Glands

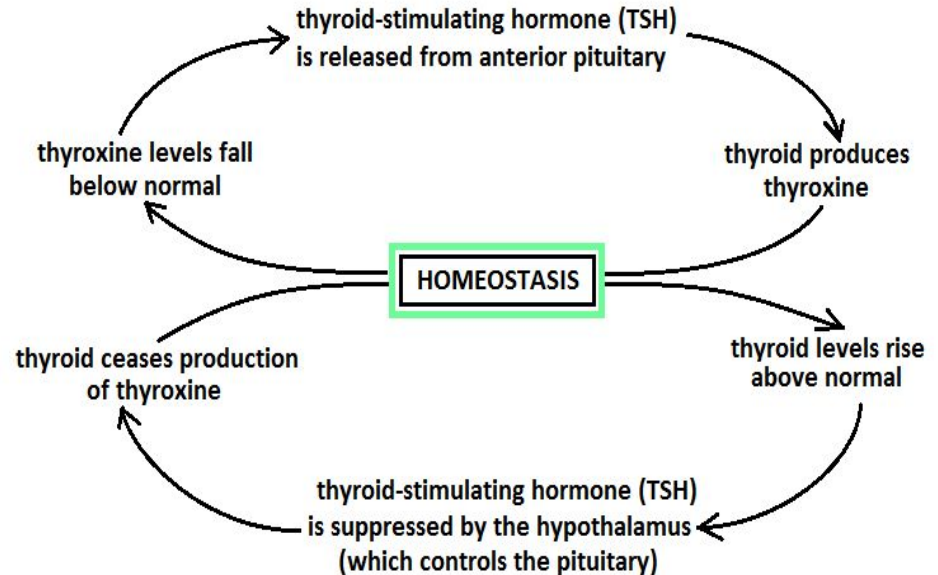


Hormones of the Thyroid and Parathyroid Glands

Thyroxine (T4)

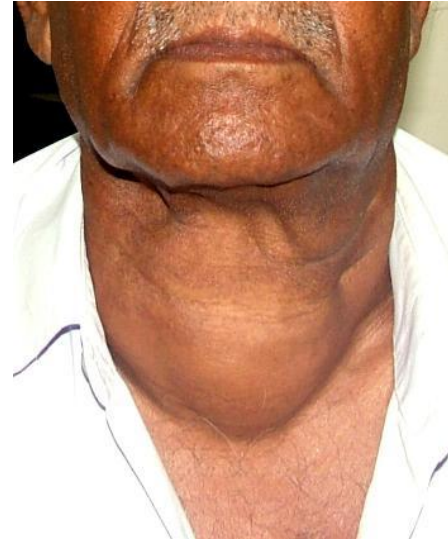
- o Released by the thyroid gland upon stimulation by TSH from the anterior pituitary
- o Its role is to increase metabolism (using up blood sugar, burning fat, increasing temperature, etc.)
- o Systemic effects (no single target)

Negative Feedback Control of Thyroxine Production



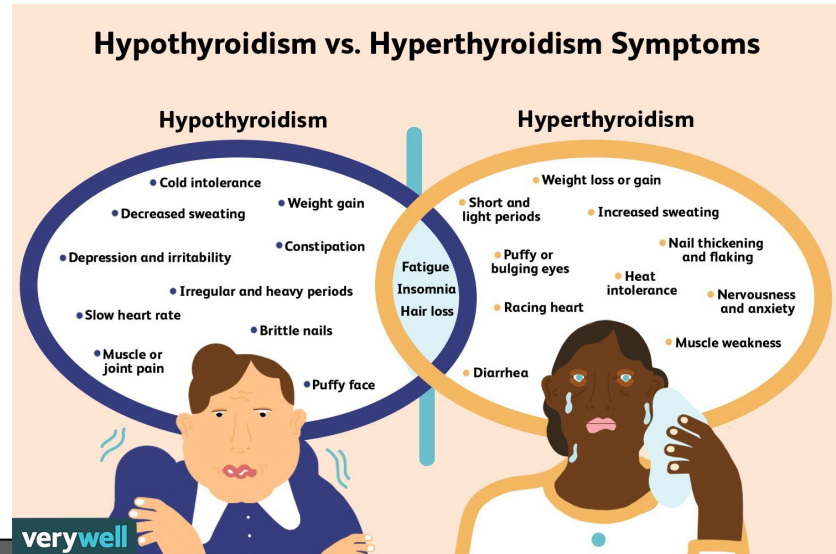
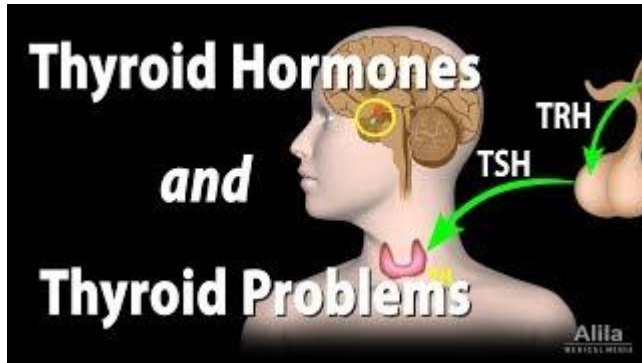
Thyroxine and Iodine

- The thyroid requires **iodine** to produce hormones such as thyroxine
- Insufficient iodine in the diet thus inhibits thyroxine production, meaning there will be no signal to stop the secretion of TSH
- The constant stimulation of the thyroid gland by TSH that results leads to the development of a goitre



Problems with Thyroxine Production

- **Hypothyroidism** is a condition where **too little thyroxine** is produced.
- **Hyperthyroidism** is where **too much thyroxine** is produced.
- This leads to opposite symptoms in affected individuals.



Problems with Thyroxine Production

This is a good place for us to pause and see how well we understand feedback loops...

Consider the following situations and make predictions:

1. A patient complains of persistent fatigue, feeling cold and weight gain. Their doctors confirms via blood test that they have hypothyroidism. What would their thyroxine and TSH levels be on a blood test?

Low thyroxine, but high TSH. Patient is not at homeostasis, and since their thyroxine levels are low, there is a constant release of TSH trying to stimulate the thyroid gland to produce more thyroxine.

Problems with Thyroxine Production

2. Let's consider a different patient who had their thyroid gland removed due to cancer. What would you expect their TSH levels to be?

Their TSH levels would be high again. There is no thyroxine produced at all, but the hypothalamus/pituitary gland do not know that there is no thyroid gland, so they continue to try to stimulate it anyways.

3. A third patient has a pituitary tumor that is leading to an overproduction of TSH. How would this impact their levels thyroxine and TRH? What symptoms would you expect?

Thyroxine would increase, and therefore we would observe symptoms of hyperthyroidism. TRH levels would be below normal.

Hormones of the Thyroid and Parathyroid Glands

Calcitonin

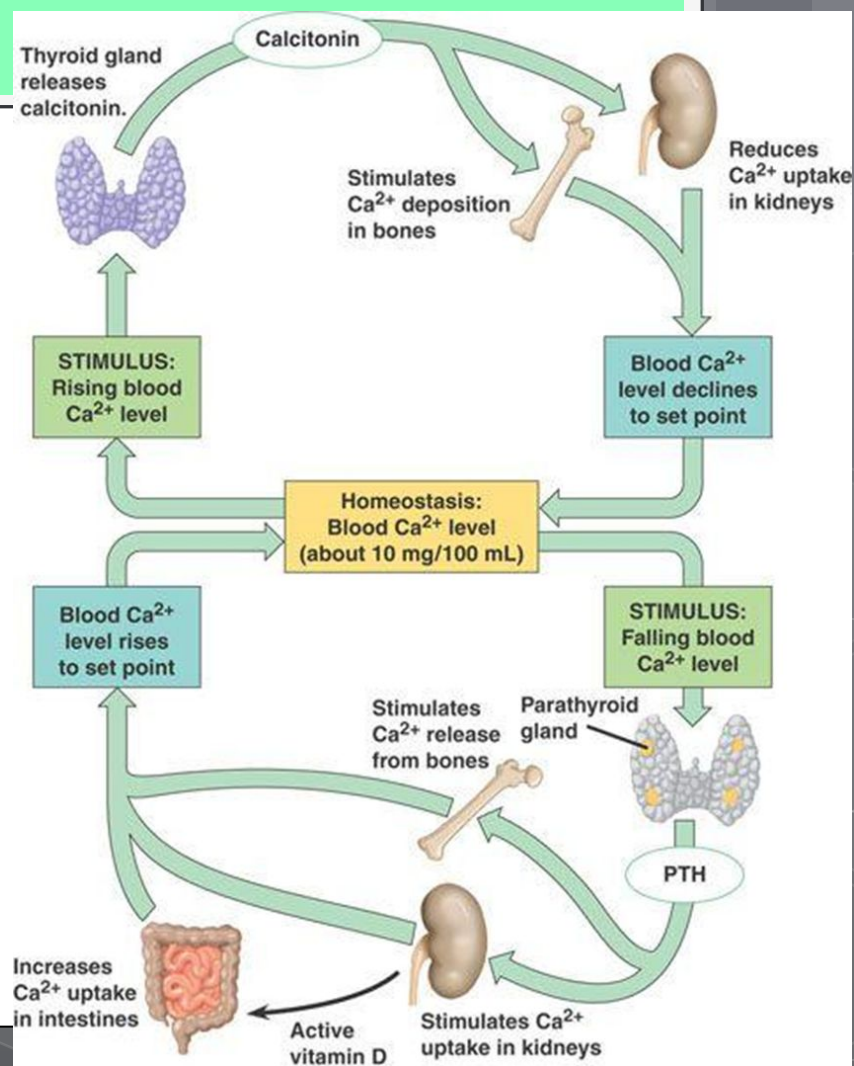
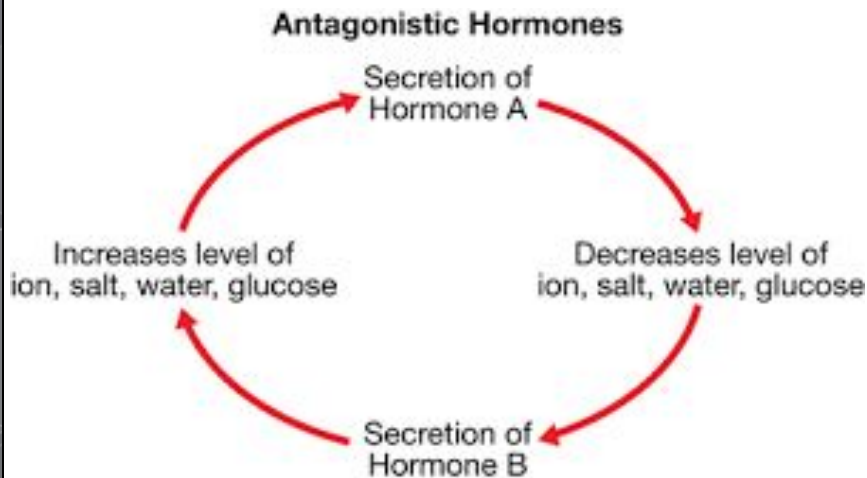
- Released by the **thyroid gland** as a response to high blood calcium (Ca^{2+}) levels
- To reduce blood calcium levels, calcitonin
 - stimulates uptake of Ca^{2+} into bones
 - reduces reabsorption of Ca^{2+} in kidneys

Parathyroid Hormone (PTH)

- Released by the **parathyroid gland** as a response to low blood calcium (Ca^{2+}) levels
- To increase blood Ca^{2+} levels, PTH
 - stimulates release of Ca^{2+} from bones
 - increases reabsorption of Ca^{2+} in kidneys

Notice anything?

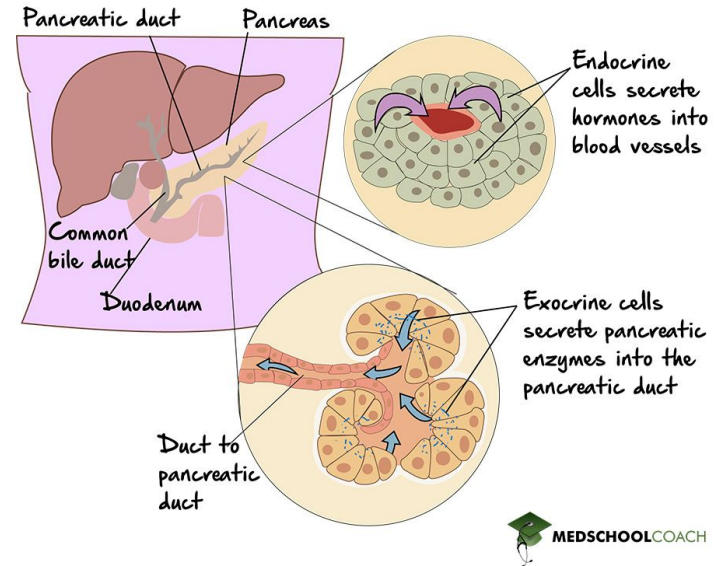
Calcitonin and PTH are an example of **antagonistic hormones** , which are a pair of hormones that have **opposing effects** .



Hormones of the Pancreas

Endocrine and Exocrine Roles of the Pancreas

- The pancreas is both an exocrine and an endocrine gland.
- In Bio 20, you learned about the exocrine functions of the pancreas. In Bio 30, we will only be interested in its endocrine functions



Hormones of the Pancreas



Insulin

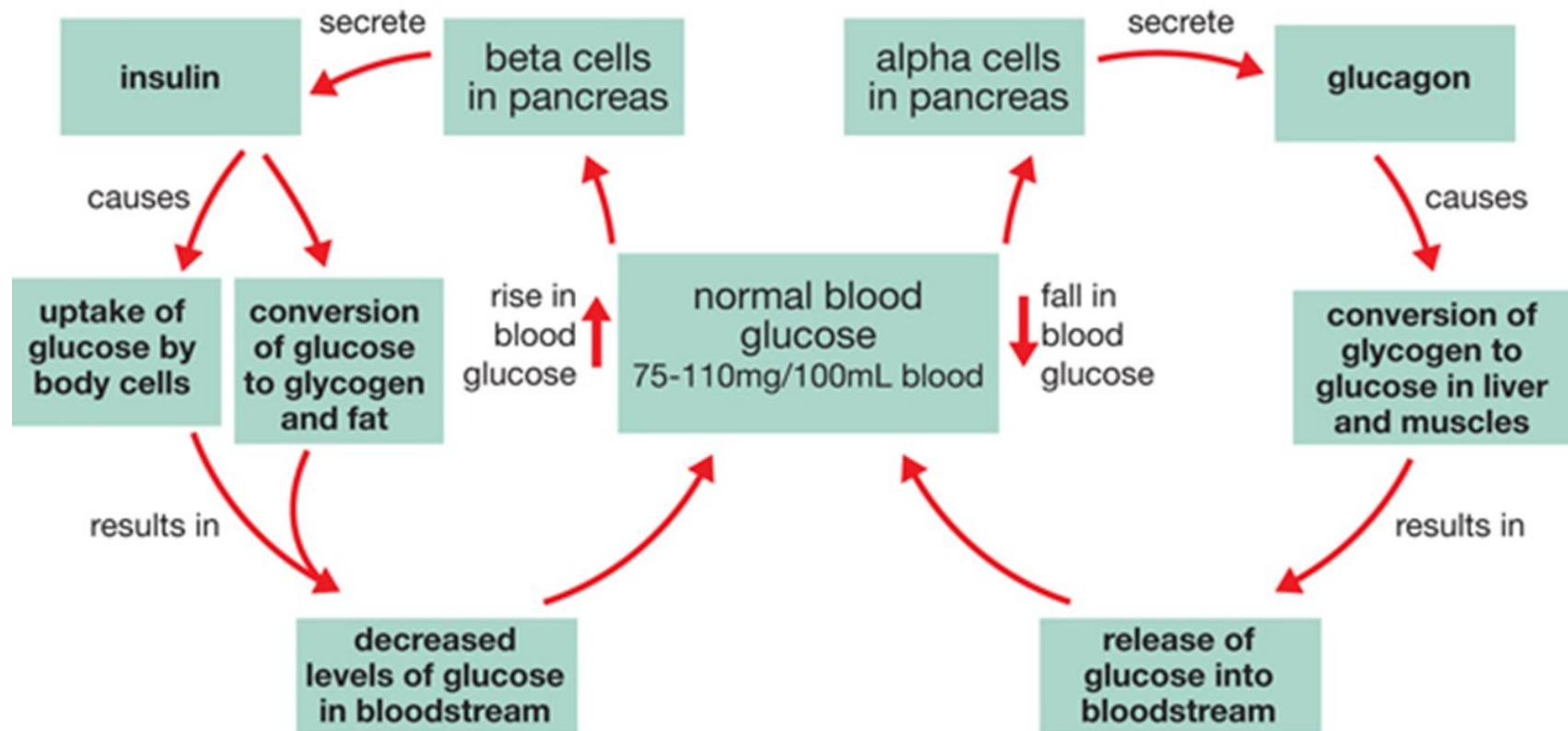
- o Released by the **beta cells of the pancreas**
- o Released after a meal when **blood sugar levels are high**
- o Makes cells more permeable to glucose, and therefore causes **cells to take in glucose** → this in turn **reduces blood sugar levels** to the normal range
- o Promotes **storage of glucose as glycogen** in the liver

Glucagon

- o Released by the **alpha cells of the pancreas**
- o Released when **blood sugar levels are low**
- o Promotes the **breakdown of glycogen in the liver to re-release glucose** into the bloodstream → this in turn **increases blood sugar levels** to the normal range

Another antagonistic hormone pair!

Key to Blood Glucose Regulation Through Negative Feedback



Diabetes Mellitus

The inability to produce enough insulin or respond appropriately to insulin

- Causes **blood glucose levels to rise sharply after meals** (hyperglycemia)
- **Cells cannot properly obtain the glucose they require for metabolism** ; individuals become fatigued
- Body compensates by metabolising fats and proteins instead of glucose. These are more difficult to break down, and release ketones, a toxic by-product.
- **Excess glucose in the bloodstream is excreted through urine ; high amounts of urine** are produced due to changes in osmotic gradient (can lead to frequent thirst)



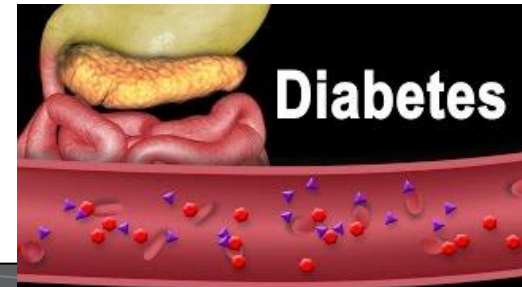
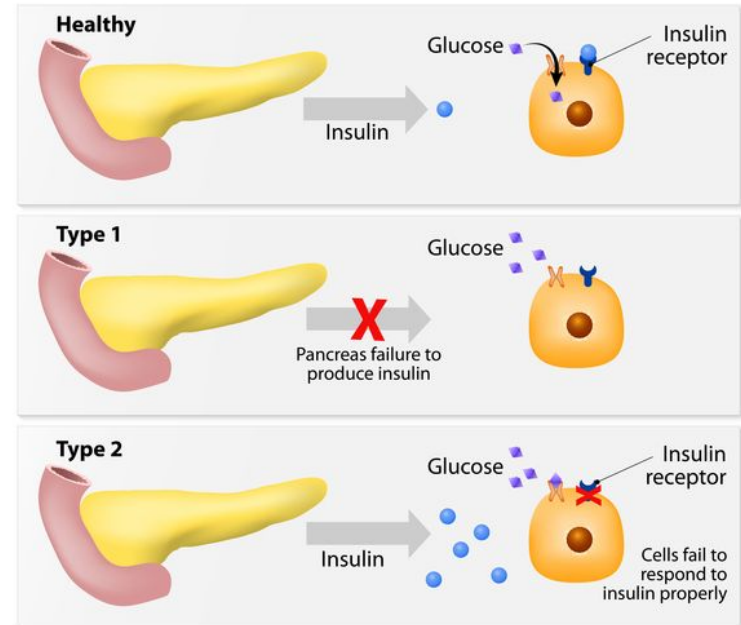
TYPE 1 DIABETES MELLITUS

- Diagnosed in childhood
- Immune system produces antibodies that attack and destroy the beta cells of the pancreas
- Leads to decreased insulin production
- Treated with daily insulin injections

TYPE 2 DIABETES MELLITUS

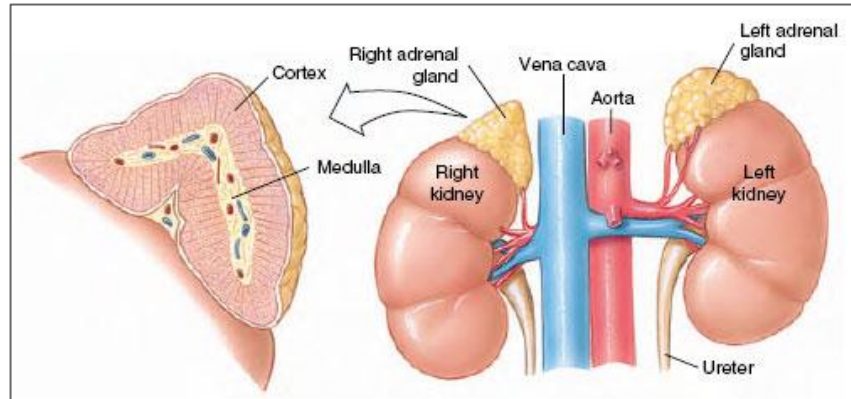
- Develops over time
- Cell receptors stop responding well to insulin
- Poor diet and lack of exercise can be contributing factors
- Treated with either insulin injections before meals OR diet, exercise and sulfonamides

DIABETES MELLITUS

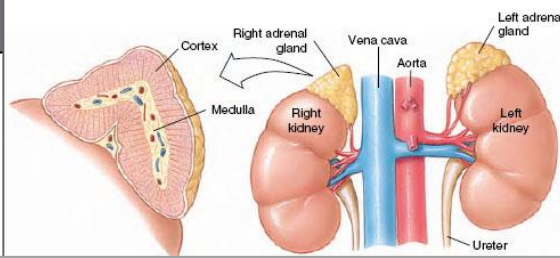


Adrenal Cortex & Adrenal Medulla

The adrenal gland is located on the top of the kidneys, and is divided into two regions: the outer **cortex** and the middle **medulla**.



- Adrenal medulla produces epinephrine and norepinephrine, involved in the short-term stress response
- Adrenal cortex produces glucocorticoids (e.g. cortisol) and mineralocorticoids (e.g. aldosterone), involved in the long-term stress response



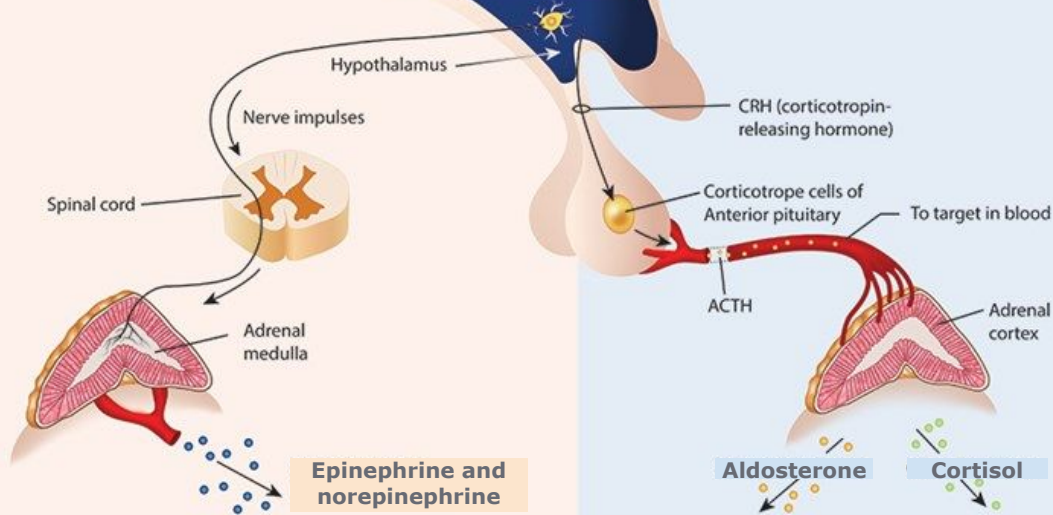
	Adrenal Cortex	Adrenal Medulla
Plays part in ...	long term stress response	short term stress response
Releases hormones ...	upon stimulation by ACTH (Adrenocorticotrophic hormone) → slower response	upon direct stimulation by nervous system → fast response
Hormones...	Cortisol: reduces pain, increases inflammation, increases energy availability by decreasing glucose storage, and by breaking down proteins Aldosterone: Increases reabsorption of Na^+ in the kidneys, which in turn increases water reabsorption leading to increased blood volume and pressure	Epinephrine and norepinephrine: initiate the fight-or-flight response (increase HR, increase breathing rate, etc.)

Symptho-Adrenomedulary System

STRESS

Internal/External
Real or Perceived

The HPA Axis



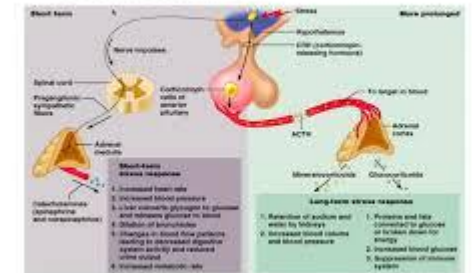
Short-term stress response

- Increased heart rate
- Increased blood pressure
- Liver converts glycogen to glucose and releases glucose to blood
- Dilatation of bronchioles
- Changes in blood flow patterns leading to increased alertness, decreased digestive system activity, and reduced urine output
- Increased metabolic rate

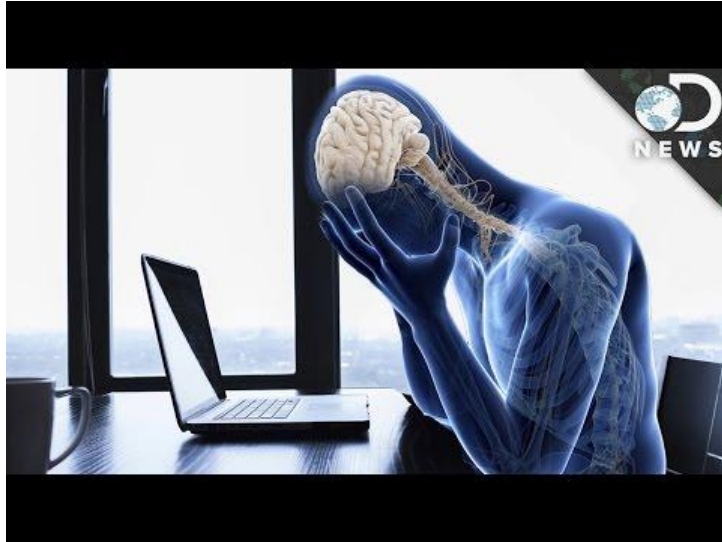
Long-term stress response

- Retention of sodium and water by kidneys
- Increased blood volume and blood pressure
- Gluconeogenesis
- Insulin Sensitivity
- GH ↑ T3
- Immune/Inflammatory Response
- Fat & Protein Mobilization

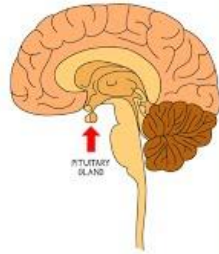
Stress and the Adrenal Gland



How does stress affect your body?



Putting it all together...



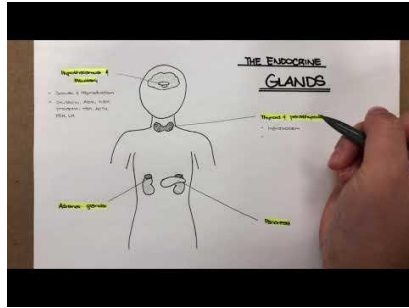
THE
PITUITARY
GLAND



THE
ANTERIOR
PITUITARY

Chapter Review:

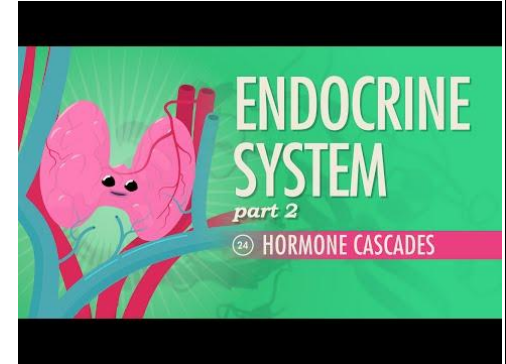
Endocrine Glands



Stress Hormones

HORMONES: RESPONSE TO STRESS

Hormone	Where it is secreted	Target organs/tissues/cells	What it does
Epinephrine	Adrenal glands	Cardiac & skeletal muscle, blood vessels	
Cortisol	Adrenal glands		
ACTH			
Adrenocortisol			



Blood Glucose Hormones

HORMONES: BLOOD GLUCOSE & WATER BALANCE

Hormone	Where it is secreted	Target organs/tissues/cells	What it does
Insulin	Pancreas	All cells except liver	Increases glucose uptake by all cells except liver & blood glucose
Glucagon	Pancreas	Liver	
ADH			

Growth & Metabolism Hormones

GH	Pituitary	Stimulates growth & metabolism during childhood	Adipose tissue
Thyroxine	Thyroid	Increases metabolic rate	Thyroid gland
TSH		Stimulates release of thyroxine	
Calcitonin			
PTH			